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A systematic review of strategies adopted to scale up COVID-19 testing in low-middle and high-income countries

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Objective: We undertook a systematic review of approaches adopted to scale up COVID-19 testing in countries across income levels.

Methods: Original studies and reports describing the strategies used to increase COVID-19 testing in countries were reviewed. The search was performed using online resources and methods while leveraging PRISMA-P guidelines. Deductive analysis to allocate relevant text from the reviewed publications/reports to the a priori themes was done.

Main results: The review covered 32 countries, including 11 high-income, two upper-middle-income,13 lower-middle-income and six low-income-countries. Most low- and middle-income countries (LMICs) increased the number of laboratories available for testing and deployed sample collection and shipment to the available laboratories. The high-income countries (HICs) i.e., South Korea, Germany, Singapore and USA developed molecular diagnostics with accompanying regulatory and legislative framework adjustments to ensure rapid development and use of the tests. HICs like South Korea leveraged existing systems to develop tests, while the LMICs leveraged existing national disease control programmes (HIV, TB, Malaria) to increase testing. Continent-wide, Africa CDC-led collaborations increased testing across most African countries by building capacity through providing testing kits and training.

Conclusion: Strategies taken appear to reflect the existing systems or economies of scale that a particular country could leverage. LMICs, for example, drew upon the infectious disease control programmes already in place to harness expertise and laboratory capacity for COVID-19 testing.

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Article summary: Strengths and limitations

- There have been challenges in testing in both developed and developing countries.
- The strength of this systematic review is that it covers many countries across different income categories and provides strategies that worked for countries that other countries could leverage to increase testing now and for future pandemics.
- The findings point to the diverse nature of the laboratory response across both developed and developing nations, mainly driven by the nature of existing capabilities and know-how.
- The major limitation of this systematic review in inability to link the testing strategies to testing rates in each country.

Introduction

At the end of 2019, a disease that presented with pneumonia in the Chinese city of Wuhan emerged.[1] The disease (caused by the SARS-CoV-2 virus) was later named COVID-19 and subsequently declared a global pandemic by the World Health Organisation (WHO) on 11th March 2020. Almost two years into the pandemic, both major and minor economies have been hugely affected, and countries will likely continue to suffer for many years after the pandemic's peak. As of 16th December 2021, a total of 273,019,599 cases and 5,338,972 deaths from COVID-19 have been reported.[2] In North America, the United States has notified most cases per 100,000 population while the United Kingdom, India, Brazil, South Africa top the list in Europe, Asia, South America and Africa, respectively.[2]

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Initially, scientists knew little about the disease and the virus, but there have been significant scientific advancements regarding prevention, diagnosis, treatment, and prevention. The world moved from literally no diagnostic tools to a plethora of diagnostic tools, which have since become available. Furthermore, there has been an unprecedented rapid development and deployment of vaccines, which could be game-changing to ending the pandemic. However, control efforts have been complicated by the emergence of variants, threatening vaccine efficacy.[3]

Beyond the non-pharmacological interventions deployed by countries to combat the pandemic, testing has been deployed to control the spread of COVID-19 by identifying, isolating and treating those infected. Indeed, a few months into the pandemic, the WHO called on countries to "test, test and test".[4,5] This was premised on the belief that early testing and contact tracing could interrupt the transmission cycle.

Amidst the calls for increased testing, WHO developed an emergency use listing procedure to expedite in vitro diagnostics to support testing,[6] to ensure the availability of diagnostics needed for the public health emergency. The WHO also made several recommendations to guide countries on testing. Initial guidance on approaches to testing was based on the number of cases reported per country (sporadic cases, clusters of cases and community transmission). The WHO suggested the need for prioritised testing strategies within countries due to the variation in levels of community transmission.[7] With the emergence of new tools like antigen detection rapid diagnostic tests (RDT), the recommendations have been updated. WHO maintained nucleic acid amplification tests (NAAT) as the reference standard and recommended antigen RDTs as a complementary strategy.[8]

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Despite the advice that testing should underpin the disease control efforts, there have been shortfalls across different countries. There is a noticeable lack of harmonised testing strategies across the globe due to several challenges, including the pre-pandemic laboratory capacity, barriers to access of tests due to increased demand, and challenges with supply. [9] To overcome some of the challenges, countries adopted different strategies to increase the number of tests per capita.

Developing countries (low- and lower-middle-income countries, see results section for definitions) were deemed more likely to bear the heaviest burden of the pandemic, mainly due to fragile health care systems, political instability, economic vulnerability, limited fiscal space, equipment shortages, among others.[10,11] This, however, does not seem to have been the case. For example, the number of cases in most African countries remains reasonably low, with low mortality rates reported.[11,12] While there are many possible reasons why cases have stayed lower than anticipated, one school of thought is that there isn't adequate testing in most of these countries. Indeed, low test per capita rates have been noted in Africa.[13] Some reports have pointed to undercounting of deaths due to COVID-19.[14]

Challenges with testing have also been observed in developed countries,[15,16] indicating that finances alone are not enough to achieve the desired testing capabilities. While the more affluent countries have outcompeted the rest on the COVID-19 diagnostics market,[17] many developing countries have used what they have at their disposal to scale-up testing.

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Future pandemics are always a possibility, and lessons must be learned from the current pandemic to enable robust and expedited responses. There is an opportunity for countries at all income levels to learn lessons from each other. Based on this background, we undertook a systematic review of strategies adopted to scale up COVID-19 testing in countries of all income levels to identify successful approaches and facilitate shared learning.

Methods

The overall aim of the systematic review was to examine strategies adopted to scale up COVID-19 testing in low-, lower-middle-, upper-middle- and high-income-countries and identify drivers underlying the selection of strategy in a given setting.

Inclusion/Exclusion criteria

We included original studies and reports that described strategies used to increase COVID-19 testing. We included articles and reports published in English between December 2019 and September 2021. We excluded any publications on research studies done, but not translated into policy and practice by the countries.

Search strategy

The search was performed using resources and methods that ensure methodological transparency and reproducibility of the findings, leveraging the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA-P) guidelines.[18] We were guided by the patient/population, intervention, comparison and outcomes (PICO) approach and stated the research question as: *What strategies were adopted to scale up COVID-19 testing in Low- Middle and High-Income countries?* To obtain more relevant literature, we reviewed the reference section of the included papers.

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We used the following terms/statements to search the studies for consideration in the systematic review: COVID-19 diagnostics in Asia, South America, Latin America, Europe, North America, COVID-19 diagnostics in high-income countries, COVID-19 diagnostics in middle-income countries, COVID-19 diagnostics in low-income countries, COVID-19 testing in high-income countries, COVID-19 diagnostics in Asia, South America, Latin America, Europe, North America, Increasing COVID-19 testing in Asia, South America, Latin America, Europe, North America, and COVID-19 laboratory testing strategies.

Data access and management

The investigation team accessed the internet sources and the relevant literature and retrieved it. The team used the Mendeley referencing manager to store, organise and manage all the references.

Database/sources searched

The researchers independently searched for studies, reports, and literature in PubMed, Google scholar, and google search engine. Titles and abstracts were independently screened by researchers based on the inclusion criteria. In addition, the quality of all the included studies and reports was assessed using predefined criteria.

Study selection criteria and process

All qualitative or quantitative studies and reports describing strategies adopted were included in the general screen. Further refinement considered only studies that offered information on the strategy used to increase COVID-19 testing in low and middle-income countries, and high-income countries were considered further for inclusion.

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The selection was carried out by the three researchers working independently. They then convened and went through the individual lists and harmonised to agree on one list of studies to be included. The three researchers have postgraduate qualifications in health-related disciplines.

Data abstraction

Both abstracts and full texts were reviewed to identify methods used to increase COVID-19 testing. A predeveloped codebook guided the identification of methods with themes and narratives of what the researchers were to consider as methods of increasing testing. Where the researchers found something not captured by the codebook, they noted this. The findings were later harmonised and summarised in a table that included authors, source of publication, date of publication, country, income category and strategy or strategies used.

Data analysis

Analysis entailed appraising and synthesising texts from the publications and reports, which were then organised into major themes and content analysis, guided by the predeveloped codebook. We used deductive analysis to allocate relevant text from the reviewed publications and reports to the a priori themes. In the process of allocating text, there were emergent new themes.

Patient and Public Involvement

Patients and the public were not involved in any way

Ethics statement

No human participants were included.

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Results

Study selection and characteristics

A total of 134 publications and 40 reports were identified by the search strategy, of which 59 peer-reviewed publications and 12 reports met the inclusion criteria and were reviewed in detail. The flow diagram (Figure 1) summarises the steps to identify the publications and reports included in the review. Supplementary table 1 represents the details of all papers and reports included and the findings.

Geo-economic coverage of included publications/reports

The selected publications and reports covered six continents (Asia, Africa, North America, Europe, South America, Oceania). There are 32 countries, and of these, 11 are high-income-countries, two are upper-middle-income, 13 are lower-middle-income, and 6 are low-income-countries. Table 1 shows the countries' corresponding Gross Domestic Product (GDP) per capita (USD).[19]

Table 1: Countries included in the systematic review and the corresponding GDP

Country	GDP (USD)
High income countries	
Canada	43241.6
France	38625.1
Belgium	44594.4
Uruguay	15438.4
Slovakia	19156.9
South Korea	31489.1
Australia	51812.2

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Chile	13231.7
USA	63543.6
Singapore	59797.8
Israel	43610.5
Upper middle-income countries	
China	10434.8
South Africa	5090.7
Peru	6126.9
Lower middle-income countries	
India	1900.7
Nepal	1155.1
Nigeria	2097.1
Vietnam	2785.7
Bangladesh	1968.8
Ghana	2328.5
Pakistan	1193.7
Mongolia	4007.3
Zambia	1050.9
Zimbabwe	1128.2
Philippines	3298.8
Bhutan	3122.4
Low-income countries	
Malawi	625.3
DRC	556.8
Rwanda	797.9
Ethiopia	936.3
Sierra Leone	484.5
Uganda	817.0

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Testing strategies applied by different countries

Through content analysis, the strategies adopted by the countries were identified. These are indicated in table 2 and summarised by income category in table 3. Notably, the strategies used were not exclusive, as some countries combined several strategies.

Table 2: Thematic analysis of the different strategies used by the countries

Theme	Strategies
Increasing accessibility	- Increase in no. of labs[9,20,21,27–29,34,39–41,45,47,54,55,59–65]

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	- Health technology assessments[21] - Setting up mobile testing units[27,38,42,62,66] - Population-wide screening/mass testing[29,67,68] - Gradual expansion of categories of people to test[5,24,27,69] - Testing only symptomatic and high priority groups[61,64–66] - Drive through testing/walk-in testing[70–75] - Community-wide testing using CHWs[48,76–78] - Sample collection and referral of samples[48] - Targeted/hot spot with mass testing[28,79] - Free COVID testing [27,29,68]
Expanding and diversifying diagnostic platforms	- Testing strategy to incorporate use of RDTs [10,21,28,34,80,81] - Development of molecular tests[23,32,59,68] - Development and subsequent use of RDTs [5,21,27,33,44,45,68] - Repurposing manufacturing capacity[30] - Massive testing with RDTs[82] - Initiatives to support production of diagnostics[31,83]
Streamlining regulatory & legislative landscape	- Licensing private providers/labs in testing[5,22,27,38–42,47,65] - Allowing research and academic institutions[5,10,41,44,46,47,60,65] - Legislation to allow for development of tests[30,65,68]
Leveraging existing systems	- Leveraging existing platforms for testing diseases [40,41,47,49,52,54,60,65,84] - Leveraging infrastructure of other disease programs[20,40,42,47,48,54,85,86] - Leveraging previous systems to develop tests [32,83] - Use of veterinary labs[41,63]
Human resource (HR) enhancement	- Training of laboratory workers [46,47,49,54,55,59,61,62] - Hiring of more personnel [29,48,54]
Creative use of technology & innovation	- Pooled testing[9,28,33] - Information technology (IT) to do surveillance and contact tracing[33,74] - Drone technology to ship samples[33,42,87]

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Leveraging regional and continental networks	- Africa CDC, PAHO in training and provision of supplies [11,47,52,61,88,89]

Table 3: Strategies by income country

	Countries using the strategy (%)			
Strategy used	**HIC**	**UMIC**	**LMIC**	**LIC**
Increase in number of labs	25.0	66.7	50.0	57.1
Health technology assessments (HTA)	0.0	0.0	8.3	0.0
Testing strategy to incorporate use of RDTs	0.0	33.3	16.7	0.0
Licensing private providers/labs in R&D and testing	37.5	66.7	25.0	14.3
Inclusion of research and academic institutions	31.3	66.7	16.7	28.6
Setting up mobile testing units	6.3	33.3	25.0	0.0
Leveraging existing testing platforms for other diseases	0.0	66.7	41.7	42.9
Development of molecular tests	31.3	0.0	8.3	0.0
Development of RDTs	25.0	0.0	33.3	0.0
Repurposing manufacturing capacity	6.3	0.0	0.0	0.0
Training of lab personnel	0.0	0.0	33.3	14.3
Increased human resource	0.0	33.3	16.7	14.3
Legislation to allow for development of tests	18.8	33.3	0.0	0.0
Leveraging existing infrastructure and systems	6.3	33.3	16.7	42.9
Expanding categories of people to test	18.8	0.0	50.0	0.0
Prioritisation of who to test (symptomatic & high priority groups	12.5	33.3	16.7	14.3
Pooled testing	6.3	0.0	8.3	0.0
Drive through testing/walk in testing	18.8	0.0	0.0	0.0
Community wide testing using CHWs	0.0	33.3	8.3	0.0
Sample collection and referral	0.0	0.0	8.3	28.6
Use of existing veterinary labs	6.3	33.3	0.0	0.0
Massive testing with RDTs	6.3	33.3	0.0	0.0
Identification of hotspots and mass testing	12.5	33.3	0.0	0.0
Initiatives to support production of diagnostics	6.3	0.0	0.0	0.0
Free COVID testing	6.3	0.0	8.3	0.0
IT to do surveillance and contact tracing	6.3	0.0	8.3	0.0
Drone technology to ship samples	0.0	0.0	8.3	0.0
Massive/large scale testing	6.3	0.0	0.0	0.0

HIC=High Income; UMIC=Upper Middle Income; LMIC=Lower middle income; LIC=Low income

The results are presented in the six thematic areas:

(i) Increasing accessibility

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Accessibility encapsulates the strategies countries adopted to ensure the testing services had a wider geographical reach closer to the people and that whoever needed a test would be able to get it. Under this theme, most countries increased the number of laboratories available for testing, while others increased sample collection points and referral to the available laboratories. The frequency of using these strategies was high in low- and middle-income countries and low in high-income countries. The increment was gradual, with more providers being brought on board to meet the demand during the pandemic. High-income countries introduced drive through testing to bring testing closer to the people. Sample collection and referral was mainly deployed by the low income and the lower-middle-income countries, while community-wide testing using community health workers was done by the middle-income countries. The high- and upper-middle-income countries increased testing rates by identifying the COVID-19 hot spots in their countries and subsequently undertook mass testing. While the middle- and high-income countries could set up mobile testing units, this wasn't the case for the low-income countries.

(ii) Expanding and diversifying Diagnostic platforms

Under this strategy, the diversity of diagnostics and testing modalities were considered. This included strategies that incorporated the use of rapid diagnostic tests (RDT), development of molecular tests, development and subsequent use of RDTs, repurposing manufacturing capacity to develop tests, community-wide testing with RDT, and initiatives to support the production of diagnostics. Development of molecular diagnostics was undertaken in the high-income countries (South Korea, Germany, Singapore, USA), but not in the low- and middle-income countries except Vietnam, which reported producing a molecular diagnostic to increase testing. In a bid to increase testing, both high-income (South Korea, Chile, United Kingdom) and lower-

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middle-income countries (India, Bangladesh, Vietnam, Ghana) developed RDTs. Adopting rapid diagnostic tests increased testing rates in these countries. Only one country (USA) reported repurposing manufacturing capacity to manufacture testing kits. Both high- and upper-middle-income countries carried out massive testing with RDTs.

(iii) Streamlining regulatory and the legislative landscape

Streamlining the regulatory and legislative landscape evolved in two ways: Accelerating the manufacturing process and increasing the number of laboratories available for testing. Governments of high-income countries adjusted the regulatory and legislative frameworks to ensure rapid development and the use of the developed tests. For example, the USA and South Korea deployed emergency use authorisation and saved the manufacturers from bureaucratic and administrative inefficiencies associated with the approval process for the diagnostics. The governments also linked test manufacturers with the clinical bodies to help fast track the validation process. As a result, tests were readily available for use in a short time frame. In Singapore, public-private partnerships were instrumental in ensuring the massive production of tests.

Governments also made regulatory and legislative adjustments to ensure more laboratories became available for COVID-19 testing. The added laboratories included the private, research and academic laboratories. This was noted mainly in the upper-middle, lower-middle- and low-income countries. For example, France, brought private providers later in the pandemic for fear of having non harmonised reporting.

Also, as part of the regulations and strategies to attract private test developers, the high-income countries (USA, South Korea) deployed initiatives to interest private

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developed in manufacturing. Such initiatives included easing the approval processes and assurance that the governments would buy the developed kits.

(iv) Leveraging existing systems

The countries leveraged existing systems in two ways. One, using the current systems and frameworks to scale up development of diagnostics and two, using the existing systems to incorporate COVID-19 testing.

High-income countries like South Korea leveraged structures established during the previous epidemics like MERS to be able to develop tests in a short time. Low- and middle-income- countries integrated COVID-19 testing within the existing disease systems to increase testing. The countries leveraged the testing systems available for other diseases like Tuberculosis, HIV and Influenza to embark on testing for COVID-19. In most of these countries, there were existing platforms for HIV viral load monitoring and Xpert MTB/RIF testing, which were leveraged to test for COVID-19. The integration also leveraged already set up systems to increase testing. For example, Nigeria leveraged the established polio systems to ensure the collection and transport of samples to the testing facilities.[20]

(v) Human resource enhancement

Countries undertook human resource initiatives. Such initiatives included recruiting more health workers and increasing their capacity through training. Recruitment of more personnel happened mainly in the upper-middle-income countries, followed by lower-middle-income countries and then the low-income countries. The trend was slightly different when it came to increasing capacity through training. The training was conducted mainly in lower-middle- and low-income countries.

(vi) Creative use of technology and innovation

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Countries such as South Korea and Ghana leveraged technology to increase COVID-19 testing. South Korea was the first country to roll out a contact tracing and surveillance application to help increase testing. Ghana reported using a contact tracing application to increase testing. Ghana also increased testing by using drones to transport samples from the field to the testing facilities established. Furthermore, Ghana innovatively used pooled testing to increase the number of samples tested with limited kits. Only India reported undertaking health technology assessment (HTA) on a locally developed COVID-19 diagnostic, whereby they assessed the impact of a rapid diagnostic test on health systems in dimensions of efficiency, equity and quality guided by the HTA framework.[21]

(vii) Leveraging regional and continental networks

Whereas most of the interventions happened at the country level, some of the efforts resulted from continent-wide collaborations led by the continental bodies. By far, the most extensive continental-led testing response was on the African continent, where the Africa CDC increased testing across most African countries by building capacity through the provision of testing kits and training. In addition, through the Africa Task Force for Corona Virus (AFCOR) and the Partnership to Accelerate COVID-19 testing (PACT), the Africa CDC helped increase testing. The Pan African Health Organisation (PAHO) led the efforts in South America and North America by ordering and providing test kits, conducting training and leveraging the national influenza centres to increase the laboratory network.

Discussion

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This systematic review of strategies adopted to scale up COVID-19 testing has revealed strategies unique to some and common to all countries. We note that the choice of strategy was mainly driven by the previous outbreak management system to leverage the country's income and manufacturing capacity. The adopted strategies fall under six thematic areas: increasing accessibility, expanding and diversifying diagnostic platforms, streamlining regulatory and legislative landscape, leveraging existing systems, human resource enhancement, creative use of technology and innovation and leveraging regional and continental networks.

The COVID-19 pandemic caught many countries unawares, irrespective of their income status. For example, in the USA and the UK, initial testing responses are marked by the absence of a clear testing strategy, defective tests, and sluggish inclusion of private laboratories.[22,23] These obstacles were later overcome, and private and academic institutions were included and expanded the categories of people prioritised for testing.[22]

Countries in a bid to avail tests to their people, took on strategies to improve access. The majority of countries gradually increased the number of testing laboratories. This was largely done in the low- and middle-income countries and to a lesser extent in high-income countries. The high-income countries introduced drive-through/walk-in testing. The lower- and upper-middle-income countries undertook community wide testing using community health care workers, a strategy supported by the decentralised nature of the health care system in most low- and middle-income countries. Countries like Zambia, South Africa used community health workers with good results.[50]

In a bid to save resources, countries adopted the symptomatologic approach to ration testing to only those exhibiting COVID-19 like symptoms. While this may have saved resources, it most likely missed many asymptomatic cases that are now thought to have driven the transmission of COVID-19. Both high-, middle- and low-income countries applied the approach, and gradually expanded the categories of individuals to test. HICs expanded testing faster than LMICs though it is important to note that apart from contact tracing, test seeking is still largely driven by experience of symptoms. Despite health care workers being at the frontline and the need to have

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them tested, the United Kingdom did not prioritise them until later in the pandemic.[51] Whereas large economies like South Korea and Singapore opened up testing to everyone and did mass testing, several African, Asian and European countries across all income categories limited testing to the symptomatic and the high priority groups, and only expanded the categories later. This could largely have been due to the logistical challenges and the need to use the supplies available sparingly as it was hard for most of them to compete.[20,44,50–53]

Countries sought to expand and diversify the diagnostic platforms available for COVID-19 testing largely through the development and/or the adoption of molecular and rapid diagnostic tests. For the high-income countries, this was achieved through the development of molecular and rapid diagnostic tests, while for other countries, it was mainly through rapid diagnostic tests. All the molecular (RT-PCR) diagnostics were developed in the high-income countries[42,58,59,62,65,66] and this in a way cushioned them from the absence of RT-PCR testing supplies that other countries were grappling with. RT-PCR testing required highly specialised laboratories that were few in most low- and middle-income countries. Some of these countries adopted rapid diagnostic tests (RDTs) in their testing algorithms as an interim measure. These were either imported or made from within. Some lower- and middle-income countries developed RDTs, and these helped increase testing.[18,32] and were gradually introduced in other low-income countries to boost testing.[9,19,52]

Some countries enhanced their regulatory and legislative landscape to maximise the number of tests within the shortest timeframe. HICs led the introduction of legislation mainly to increase manufacturing capacity, adoption of new diagnostic tests and accreditation of non-traditional facilities for involvement in testing COVID-19. In the USA, legislation to modify the regulatory framework governing laboratory developed tests led to more tests being introduced later.[33] However, this came several months into the pandemic. The federal government was blamed for the red tape that squeezed out academic and private laboratories in testing and product development.[35] In an emergency, time is of essence and lifting non-critical administrative barriers without compromising oversight can help expedite the delivery of vital products like diagnostics.[36] As part of regulation and licencing, countries included private, academic and research laboratories in the response.

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There were differences in the engagement of private players between high-income countries and the low- and middle-income countries and the point at which they were brought on board. While the high-income countries private sector was engaged in the early days of the pandemic as test developers [22,31,37], in the low- and middle-income countries, they were engaged in testing.[38–43] In most of the high-income countries the private laboratories were engaged in testing quite late.[22,26] Countries called on academic institutions to support the testing efforts. This was noted across all countries i.e., low-, middle- and high-income countries.[44–46] The academic institutions in most countries have molecular platforms they use mainly for research purposes, and when leveraged could help supplement government level efforts to increase testing. Furthermore, most academic institutions in low- and middle-income countries have collaborative links with partners in the developed world. Such networks could have been leveraged to ensure the availability of kits for testing. Early in the pandemic, the United Kingdom engaged the academic institutions by asking them to avail their platforms for use in public facilities. This was later changed and instead the academic institutions were called on to do the testing.[5]

Countries also tapped into the available systems and previous experiences to mount responses to increase COVID-19 testing. Whereas the high-income countries used prior experience to develop tests, the low-and middle-income countries leveraged the systems available to increase testing. The existing systems worked in favour of low- and middle-income-countries as they could tap into these to effect testing. The Polymerase chain reaction (PCR) testing footprint within the disease programs like TB, HIV, Influenza, Schistosomiasis were used by most of the low- and middle-income countries.[29,41,47,48] All they needed were supplies and the collaborations and donations helped with this.[11,47,50] With most of the low and middle income countries accounting for the highest burden of TB[51], there have been progressive investments in diagnostics for TB and these have been leveraged for COVID-19 testing.[52] Korea was able to rapidly develop testing kits by leveraging the test development strategy used for MERS.[53]

Human resources underpin most of the testing needed by the countries, as having equipment alone is not enough. As a result, countries recruited and trained more

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personnel.[11,22,29,48,55] Recruitment of extra personnel was mainly reported by the upper middle-income countries followed by the lower middle-income countries, while the training of personnel was reported mainly by the lower middle-income countries followed by the low-income countries. Human resource challenges in low- and middle-income countries have previously been documented[56,] and the use of this as a strategy was possibly to meet the short falls in number available to run the COVID-19 tests. The recruitment and training of personnel was not noted in any of the high-income countries, possibly a pointer that these countries are not grappling with challenges of limited human resources for health.

Some countries were creative and innovative in their approach. In order to save the limited logistics, Ghana adopted pooled testing.[33] Specimen pooling is a method of screening a large number of patients for an infection and typically involves combining multiple patient specimens into a single test tube than testing multiple samples in separate reaction tubes.[54] This approach enabled Ghana increase the number of tests done to the extent that in Africa they posted the highest number of tests done only behind South Africa.[33]

There were strategies at continental level, and these were overseen by the relevant bodies. By far, the greatest demonstration of continental led efforts was in Africa where the African centres for disease control and prevention (Africa CDC). Africa CDC demonstrated the need for harmonisation, solidarity, and coordination at a continental level. They launched a joint continental strategy, the partnership to accelerate COVID-19 testing, purchased kits and equipped several African laboratories.[11,50] African countries were outcompeted on the global supplies market and through the Africa CDC, pooled purchasing was implemented so that African countries competed as a block (as one large customer).[47] Similar efforts are noted in South America where the Pan American Health Organisation (PAHO) oversaw country level coordination, planning and monitoring. It mobilised experts and furnished countries with laboratory supplies for detection of the virus.[58]

The limitations of this review are that there might have been strategies adopted by other countries but were never published and thus didn't appear anywhere in the searched databases.

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Conclusion

The review has revealed that all countries regardless of their income level had unique challenges associated with ramping up COVID-19 testing. Whereas the low- and middle-income countries have struggled to increase the testing rates as reflected by the low testing rates, they have been able to post optimal testing rates because of experience of dealing with disease outbreaks and leveraging of the existing disease control programs. The high-income countries in Europe and North America, despite their large financial envelope, also struggled having enough tests available in the short to medium term of COVID-19 pandemic. This was not the case in Asia where the high-income economies leveraged the previous disease outbreak experience like SARS, MERS to have in place diagnostics in a short period of time and quickly put in place effective testing strategies. Country's epidemic handling experience, and manufacturing and financial landscape were critical in informing strategy to ramp up testing. Learning lessons from the COVID-19 pandemic and strengthening health systems will put the world in a better position to respond more effectively to any future pandemics.

CONTRIBUTORSHIP

WM, WS: Conceptualisation, funding acquisition, project administration, data curation, formal analysis; WM, WS, BO, VS, AR, BK, PT, BM: Investigation, methodology, manuscript revision; WM: data curation, formal analysis, original draft writing

COMPETING INTERESTS

None declared

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DATA SHARING

Data are available upon reasonable request. will be made available to others upon reasonable request. A formal request should be sent via email to the corresponding author Dr Winters Muttamba at wm55@st-andrews.ac.uk

Figure 1: Flow diagram showing selection of included peer reviewed publications and reports

References

1. Shi Y, Wang G, Cai X peng, et al. An overview of COVID-19. *Journal of Zhejiang University-SCIENCE B 2020 21:5*. 2020;21(5):343-360. doi:10.1631/JZUS.B2000083
2. COVID-19 Map - Johns Hopkins Coronavirus Resource Center. Accessed 17th December, 2021. https://coronavirus.jhu.edu/map.html
3. Tregoning JS, Flight KE, Higham SL, Wang Z, Pierce BF. Progress of the COVID-19 vaccine effort: viruses, vaccines and variants versus efficacy, effectiveness and escape. *Nature Reviews Immunology*. Published online 2021. doi:10.1038/S41577-021-00592-1
4. WHO Director-General's opening remarks at the media briefing on COVID-19 - 16th March 2020. Accessed 28th September, 2021. https://www.who.int/director-general/speeches/detail/who-director-general-s-opening-remarks-at-the-media-briefing-on-covid-19---16-march-2020
5. Rough E. Coronavirus: Testing for Covid-19. *Number CBP*. 2020;8897. Accessed September 27, 2021. www.parliament.uk/commons-library|intranet.parliament.uk/commons-library|papers@parliament.uk|@commonslibrary
6. Coronavirus disease (COVID-19) Pandemic — Emergency Use Listing Procedure (EUL) open for IVDs | WHO - Prequalification of Medical Products (IVDs, Medicines, Vaccines and Immunization Devices, Vector Control). Accessed 20th October, 2021. https://extranet.who.int/pqweb/vitro-diagnostics/coronavirus-disease-covid-19-pandemic-%E2%80%94-emergency-use-listing-procedure-eul-open
7. Laboratory testing strategy recommendations for COVID-19: Interim guidance.
8. Recommendations for national SARS-CoV-2 testing strategies and diagnostic capacities Interim guidance. Published online 2021.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

9. Benítez MA, Velasco C, Sequeira AR, Henríquez J, Menezes FM, Paolucci F. Responses to COVID-19 in five Latin American countries. *Health Policy and Technology*. 2020;9(4):525-559. doi:10.1016/J.HLPT.2020.08.014
10. Garcia PJ, Alarcón A, Bayer A, et al. COVID-19 Response in Latin America. *The American Journal of Tropical Medicine and Hygiene*. 2020;103(5):1765. doi:10.4269/AJTMH.20-0765
11. Medinilla A, Byiers B, Apiko P. ecdpm's African regional responses to COVID-19. Published online 2020. Accessed 20th September, 2021. www.ecdpm.org/dp272
12. Total confirmed COVID-19 cases. Accessed 28th September, 2021. https://ourworldindata.org/grapher/covid-cases-income
13. Hasell J, Mathieu E, Beltekian D, et al. A cross-country database of COVID-19 testing. *Scientific Data*. 2020;7(1). doi:10.1038/S41597-020-00688-8
14. A Continent Where the Dead Are Not Counted - The New York Times. Accessed 29th November, 2021. https://www.nytimes.com/2021/01/02/world/africa/africa-coronavirus-deaths-underreporting.html
15. of Health D, Care S. Coronavirus (COVID-19) Scaling up our testing programmes Department of Health and Social Care. Published online 2020.
16. FDA Emergency Use Authorization (EUA) Guidance for COVID-19 testing. Accessed 28th September, 2021. https://crsreports.congress.gov
17. Nkengasong J. Let Africa into the market for COVID-19 diagnostics. *Nature*. 2020;580(7805):565-565. doi:10.1038/D41586-020-01265-0
18. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *The BMJ*. 2021;372. doi:10.1136/BMJ.N71
19. GDP per capita (current US$) | Data. Accessed 20th October, 2021. https://data.worldbank.org/indicator/NY.GDP.PCAP.CD
20. National Strategy to Scale Up Access to Coronavirus Disease Testing in Nigeria.
21. Mukherjee K. Integrating technology, innovation and policy: COVID-19 and HTA. *Health Policy and Technology*. 2021;10(1):16-20. doi:10.1016/J.HLPT.2021.01.003
22. Alexander M, Unruh L, Koval A, Belanger W. United States response to the COVID-19 pandemic, January–November 2020. *Health Economics, Policy and Law*. Published online 2021:1-14. doi:10.1017/S1744133121000116
23. Jamison DT, Wu KB. The East–West Divide in Response to COVID-19. *Engineering*. Published online 12th June, 2021. doi:10.1016/J.ENG.2021.05.008
24. Chipimo PJ, Barradas DT, Kayeyi N, et al. First 100 Persons with COVID-19 — Zambia, March 18–April 28, 2020. *Morbidity and Mortality Weekly Report*. 2020;69(42):1547. doi:10.15585/MMWR.MM6942A5
25. of Health D, Care S. Coronavirus (COVID-19) Scaling up our testing programmes Department of Health and Social Care. Published online 2020.
26. Desson Z, Weller E, McMeekin P, Ammi M. An analysis of the policy responses to the COVID-19 pandemic in France, Belgium, and Canada. *Health Policy and Technology*. 2020;9(4):430-446. doi:10.1016/J.HLPT.2020.09.002
27. Group GC 19 S. Combating the COVID-19 pandemic in a resource-constrained setting: insights from initial response in India. *BMJ Global Health*. 2020;5(11):e003416. doi:10.1136/BMJGH-2020-003416
28. Nyazika TK, Kaela R, Mugoni M, et al. Implementation of Antibody Rapid Diagnostic Testing versus Real-Time Reverse Transcription-PCR Sample Pooling in the Screening of COVID-19: a Case of Different Testing Strategies in Africa. *mSphere*. 2020;5(4). doi:10.1128/MSPHERE.00524-20

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

29. Dorji T. The Gross National Happiness Framework and the Health System Response to the COVID-19 Pandemic in Bhutan. *The American Journal of Tropical Medicine and Hygiene*. 2021;104(2):441. doi:10.4269/AJTMH.20-1416
30. Vandenberg O, Martiny D, Rochas O, van Belkum A, Kozlakidis Z. Considerations for diagnostic COVID-19 tests. *Nature Reviews Microbiology 2020 19:3*. 2020;19(3):171-183. doi:10.1038/s41579-020-00461-z
31. Tromberg BJ, Schwetz TA, Pérez-Stable EJ, et al. Rapid Scaling Up of Covid-19 Diagnostic Testing in the United States — The NIH RADx Initiative. *https://doi.org/101056/NEJMsr2022263*. 2020;383(11):1071-1077. doi:10.1056/NEJMSR2022263
32. Kuguyo O, Kengne AP, Dandara C. Singapore COVID-19 Pandemic Response as a Successful Model Framework for Low-Resource Health Care Settings in Africa? *https://home.liebertpub.com/omi*. 2020;24(8):470-478. doi:10.1089/OMI.2020.0077
33. Sibiri H, Zankawah SM, Prah D. Coronavirus diseases 2019 (COVID-19) response: Highlights of Ghana's scientific and technological innovativeness and breakthroughs. *Ethics, Medicine and Public Health*. 2020;14:100537. doi:10.1016/J.JEMEP.2020.100537
34. Budhathoki P, Shrestha DB, Khadka S, Giri S. COVID-19 Status in Nepal and the Way Forward. *Europasian Journal of Medical Sciences* . 2020;2(2):99-104. doi:10.46405/EJMS.V2I2.75
35. How the government delayed coronavirus testing - CNNPolitics. Accessed 22nd October, 2021. https://edition.cnn.com/2020/04/09/politics/coronavirus-testing-cdc-fda-red-tape-invs/index.html
36. Note by the Secretariat in consultation with the Chairs and the Bureaus of the Regulatory Policy Committee and the Network of Economic Regulators Regulatory Quality and COVID-19: Managing the Risks and Supporting the Recovery. Published online 2020.
37. Lee D, Lee J. Testing on the move: South Korea's rapid response to the COVID-19 pandemic. *Transportation Research Interdisciplinary Perspectives*. 2020;5:100111. doi:10.1016/J.TRIP.2020.100111
38. Ogbolosingha AJ, Singh A. COVID-19 pandemic: Review of impediments to public health measures in Sub-Saharan Africa. Published online 2020. doi:10.5455/ajpmph.20200512020014
39. Siddiqui AF, Wiederkehr M, Rozanova L, Flahault A. Situation of India in the COVID-19 Pandemic: India's Initial Pandemic Experience. *International Journal of Environmental Research and Public Health 2020, Vol 17, Page 8994*. 2020;17(23):8994. doi:10.3390/IJERPH17238994
40. Al-Mustapha AI, Tijani AA, Oyewo M, et al. Nigeria's race to zero COVID-19 cases: True disease burden or testing failure? *Journal of Global Health*. 2021;11. doi:10.7189/JOGH.11.03094
41. Acheampong G, Owusu M, Nkrumah B, et al. Laboratory capacity in COVID-19 diagnosis and the need to enhance molecular testing in Ghana. *https://doi.org/101080/23779497202119081 57*. 2021;6(1):10-17. doi:10.1080/23779497.2021.1908157
42. Adepoju P. Africa's struggle with inadequate COVID-19 testing. *The Lancet Microbe*. 2020;1(1):e12. doi:10.1016/S2666-5247(20)30014-8
43. Kebede A, Lanyero B, Beyene B, et al. Expanding molecular diagnostic capacity for COVID-19 in Ethiopia: operational implications, challenges and lessons learnt. *The Pan African Medical Journal*. 2021;38:1-11. doi:10.11604/PAMJ.2021.38.68.27501

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

44. Taylor L. Uruguay is winning against covid-19. This is how. *BMJ*. 2020;370. doi:10.1136/BMJ.M3575
45. Anwar S, Nasrullah M, Hosen MJ. COVID-19 and Bangladesh: Challenges and How to Address Them. *Frontiers in Public Health*. 2020;0:154. doi:10.3389/FPUBH.2020.00154
46. Assefa N, Hassen JY, Admassu D, et al. COVID-19 Testing Experience in a Resource-Limited Setting: The Use of Existing Facilities in Public Health Emergency Management. *Frontiers in Public Health*. 2021;0:718. doi:10.3389/FPUBH.2021.675553
47. Ondoa P, Kebede Y, Loembe MM, et al. COVID-19 testing in Africa: lessons learnt. *The Lancet Microbe*. 2020;1(3):e103-e104. doi:10.1016/S2666-5247(20)30068-9
48. Nachega JB, Grimwood A, Mahomed H, et al. From Easing Lockdowns to Scaling Up Community-based Coronavirus Disease 2019 Screening, Testing, and Contact Tracing in Africa—Shared Approaches, Innovations, and Challenges to Minimise Morbidity and Mortality. *Clinical Infectious Diseases*. 2021;72(2):327-331. doi:10.1093/CID/CIAA695
49. Abera A, Belay H, Zewude A, et al. Establishment of COVID-19 testing laboratory in resource-limited settings: challenges and prospects reported from Ethiopia. *https://doi.org/101080/165497162020184196*3. 2020;13(1). doi:10.1080/16549716.2020.1841963
50. Patterson AS, Balogun E. African Responses to COVID-19: The Reckoning of Agency? *African Studies Review*. 2021;64(1):144-167. doi:10.1017/ASR.2020.122
51. Global Tuberculosis Report 2021. Accessed 22nd October, 2021. https://www.who.int/teams/global-tuberculosis-programme/tb-reports/global-tuberculosis-report-2021
52. Table of Contents.
53. South Korea's Response to COVID-19 | FDA. Accessed 22nd October, 2021. https://www.fda.gov/medical-devices/coronavirus-covid-19-and-medical-devices/south-koreas-response-covid-19
54. Saha S, Pai M. Can COVID-19 innovations and systems help low- and middle-income countries to re-imagine healthcare delivery? *Med*. 2021;2(4):369-373. doi:10.1016/J.MEDJ.2021.02.008
55. Nawaz A, Su X, Barkat MQ, et al. Epidemic Spread and Its Management Through Governance and Leadership Response Influencing the Arising Challenges Around COVID-19 in Pakistan—A Lesson Learnt for Low Income Countries With Limited Resource. *Frontiers in Public Health*. 2020;8. doi:10.3389/FPUBH.2020.573431/PDF
56. Hongoro C, McPake B. How to bridge the gap in human resources for health. *The Lancet*. 2004;364(9443):1451-1456. doi:10.1016/S0140-6736(04)17229-2
57. Wacharapluesadee S, Kaewpom T, Ampoot W, et al. Evaluating the efficiency of specimen pooling for PCR-based detection of COVID-19. *Journal of Medical Virology*. 2020;92(10):2193-2199. doi:10.1002/JMV.26005
58. Pan American Health Organization Response to COVID-19 in the Americas, 17th January to 31st May 2020 - PAHO/WHO | Pan American Health Organization. Accessed 27th September, 2021. https://www.paho.org/en/documents/pan-american-health-organization-response-covid-19-americas-17-january-31-may-2020
59. Nguyen TV, Tran QD, Phan LT, et al. In the interest of public safety: rapid response to the COVID-19 epidemic in Vietnam. *BMJ Global Health*. 2021;6(1):e004100. doi:10.1136/BMJGH-2020-004100
60. Munharo S, Nayupe SF, Patel P. Challenges of COVID-19 testing in low-middle income countries (LMICs) Challenges of COVID-19 testing in low-middle income

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countries (LMICs): the case of Malawi View project Patrick Mbulaje Centre for the Development of People (CEDEP) 4 PUBLICATIONS 3 CITATIONS SEE PROFILE. doi:10.21037/jlpm-20-84

61. Amzat J, Aminu K, Kolo VI, Akinyele AA, Ogundairo JA, Danjibo MC. Coronavirus outbreak in Nigeria: Burden and socio-medical response during the first 100 days. *International Journal of Infectious Diseases*. 2020;98:218-224. doi:10.1016/J.IJID.2020.06.067
62. Waris A, Atta UK, Ali M, Asmat A, Baset A. COVID-19 outbreak: current scenario of Pakistan. *New Microbes and New Infections*. 2020;35:100681. doi:10.1016/J.NMNI.2020.100681
63. Dhakal S, Karki S. Early Epidemiological Features of COVID-19 in Nepal and Public Health Response. *Frontiers in Medicine*. 2020;0:524. doi:10.3389/FMED.2020.00524
64. Ferree KE, Harris AS, Dulani B, Kao K, Lust E, Metheney E. Stigma, Trust, and procedural integrity: Covid-19 testing in Malawi. *World Development*. 2021;141:105351. doi:10.1016/J.WORLDDEV.2020.105351
65. K K, MM B, R C, et al. Testing COVID-19 in Brazil: fragmented efforts and challenges to expand diagnostic capacity at the Brazilian Unified National Health System. *Cadernos de saude publica*. 2021;37(3). doi:10.1590/0102-311X00277420
66. How resilient have European health systems been to the COVID-19 crisis? Published online 19th November, 2020. doi:10.1787/85E4B6A1-EN
67. Mahase E. Covid-19: Mass testing in Slovakia may have helped cut infections. *BMJ*. 2020;371:m4761. doi:10.1136/BMJ.M4761
68. Duchâtel M. ABOUT THE AUTHORS. Published online 2015. Accessed 27th September, 2021. www.institutmontaigne.org/en
69. COVID-19 Prioritisation of Diagnostic Testing.
70. Choi S, Han C, Lee J, Kim SI, Kim IB. Innovative screening tests for COVID-19 in South Korea. *Clinical and Experimental Emergency Medicine*. 2020;7(2):73. doi:10.15441/CEEM.20.032
71. Kwon KT, Ko JH, Shin H, Sung M, Kim JY. Drive-Through Screening Center for COVID-19: a Safe and Efficient Screening System against Massive Community Outbreak. *Journal of Korean Medical Science*. 2020;35(11):123. doi:10.3346/JKMS.2020.35.E123
72. Kim E. Drawing on Israel's Experience Organising Volunteers to Operationalise Drive-Through Coronavirus Testing Centers. *Disaster Medicine and Public Health Preparedness*. 2020;14(6):812-814. doi:10.1017/DMP.2020.104
73. Evans LK, Shinagawa A, Sutton S, Calvo L. COVID-19 Drive-Through Point of Screening and Testing (POST) System: A Safe, Efficient, and Adaptable Model for Nasopharyngeal Swab Collection. *Disaster Medicine and Public Health Preparedness*. Published online 2020:1-7. doi:10.1017/DMP.2020.313
74. Madhi SA, Baillie V, Cutland CL, et al. Safety and efficacy of the ChAdOx1 nCoV-19 (AZD1222) Covid-19 vaccine against the B.1.351 variant in South Africa. *Alex Sigal*. 2021;13:2. doi:10.1101/2021.02.10.21251247
75. Shah A, Challener D, Tande AJ, et al. Drive-Through Testing: A Unique, Efficient Method of Collecting Large Volume of Specimens During the SARS-CoV-2 (COVID-19) Pandemic. *Mayo Clinic Proceedings*. 2020;95(7):1420-1425. doi:10.1016/J.MAYOCP.2020.04.030
76. Odendaal N. Constructing an "infrastructure of care" – understanding the institutional remnants and socio-technical practices that constitute South Africa's Covid-19 response. *https://doi.org/101080/0272363820201807166*. 2020;42(3):391-398. doi:10.1080/02723638.2020.1807166

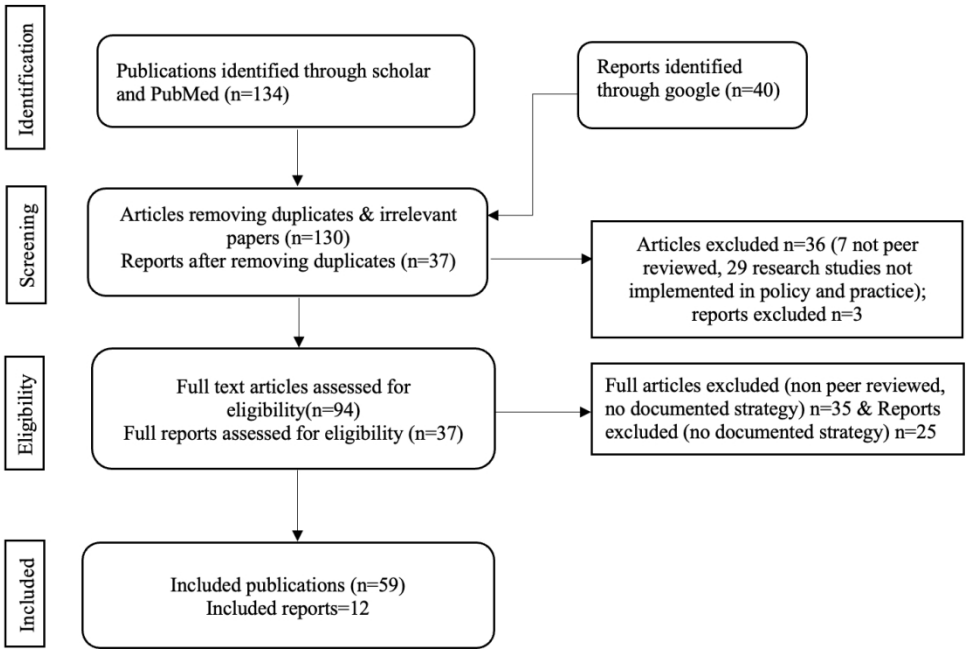
For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

77. Massinga Loembé M, Tshangela A, Salyer SJ, Varma JK, Ouma AEO, Nkengasong JN. COVID-19 in Africa: the spread and response. *Nature Medicine 2020 26:7*. 2020;26(7):999-1003. doi:10.1038/s41591-020-0961-x
78. Mendelson M, Madhi S. South Africa's coronavirus testing strategy is broken and not fit for purpose: It's time for a change. *South African Medical Journal*. 2020;110(6):429-431. doi:10.7196/SAMJ.2020.V110I6.14902
79. Boretti A. The Increasing Number of Infected in Victoria, Australia since 15th June, 2020, is the Result of Over-Testing and Over-Controlling without Safety. *Integrative Journal of Medical Sciences*. 2020;7:187. doi:10.15342/IJMS.7.187
80. Herrera-Añazco P, Uyen-Cateriano A, Mezones-Holguin E, et al. Some lessons that Peru did not learn before the second wave of COVID-19. *The International Journal of Health Planning and Management*. 2021;36(3):995-998. doi:10.1002/HPM.3135
81. Laxminarayan R, Jameel S, Sarkar S. India's Battle against COVID-19: Progress and Challenges. *The American Journal of Tropical Medicine and Hygiene*. 2020;103(4):1343. doi:10.4269/AJTMH.20-0992
82. Frnda J, Durica M. On Pilot Massive COVID-19 Testing by Antigen Tests in Europe. Case Study: Slovakia. *Infectious Disease Reports 2021, Vol 13, Pages 45-57*. 2021;13(1):45-57. doi:10.3390/IDR13010007
83. Vandenberg O, Martiny D, Rochas O, van Belkum A, Kozlakidis Z. Considerations for diagnostic COVID-19 tests. *Nature Reviews Microbiology 2020 19:3*. 2020;19(3):171-183. doi:10.1038/s41579-020-00461-z
84. Lau LL, Hung N, Wilson K. COVID-19 response strategies: considering inequalities between and within countries. *International Journal for Equity in Health 2020 19:1*. 2020;19(1):1-3. doi:10.1186/S12939-020-01254-9
85. Erkhembayar R, Dickinson E, Badarch D, et al. Early policy actions and emergency response to the COVID-19 pandemic in Mongolia: experiences and challenges. *The Lancet Global Health*. 2020;8(9):e1234-e1241. doi:10.1016/S2214-109X(20)30295-3
86. Dzinamarira T, Dzobo M, Chitungo I. COVID-19: A perspective on Africa's capacity and response. *Journal of Medical Virology*. 2020;92(11):2465-2472. doi:10.1002/JMV.26159
87. Sarfo AK, Karuppannan S. Application of Geospatial Technologies in the COVID-19 Fight of Ghana. *Transactions of the Indian National Academy of Engineering 2020 5:2*. 2020;5(2):193-204. doi:10.1007/S41403-020-00145-3
88. Kavanagh MM, Erondu NA, Tomori O, et al. Access to lifesaving medical resources for African countries: COVID-19 testing and response, ethics, and politics. *The Lancet*. 2020;395(10238):1735-1738. doi:10.1016/S0140-6736(20)31093-X
89. Organisation PAH. Pan American Health Organization Response to COVID-19 in the Americas, 17 January to 31 May 2020. Published online July 6, 2020. Accessed September 27, 2021. https://iris.paho.org/handle/10665.2/52453

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Supplementary table 1: Strategies used by each country by thematic area

	Strategy	Countries	Thematic area (6)
1	Increase in no. of labs	Bangladesh, Malawi, Nepal, Pakistan, Yemen, Nigeria, Ghana, Malawi, Peru, Ethiopia, Chile, Bhutan, UK, Uganda, Nepal, Canada, Uruguay, Vietnam, Nigeria, Brazil	Increasing accessibility
2	Health technology assessments	India	Increasing accessibility
3	Testing strategy to incorporate use of RDTs	India, Zimbabwe, Peru	Expanding and diversifying diagnostic platforms
4	Licensing private providers/labs in testing	South Africa, India, Nigeria, Ghana, Ethiopia, China, Chile, USA, UK, North Korea, Singapore, Brazil	Streamlining regulatory & legislative landscape
5	Allowing research and academic institutions	South Africa, Nigeria, France, Uruguay, Malawi, Ethiopia, Ghana, Chile, UK, USA, South Korea, Singapore, Brazil	Streamlining regulatory & legislative landscape
6	Setting up mobile testing units	India, Nigeria, South Africa, Pakistan, Lithuania	Increasing accessibility
7	Leveraging existing platforms for testing diseases	India, Malawi, South Africa, Mongolia, Ethiopia, Nigeria, Ghana, Philippines, Uganda, Brazil	Leveraging existing systems
8	Leveraging infrastructure of other disease programs	Bangladesh, South Africa, Mongolia, Nigeria, Sierra Leone, Ethiopia, Uganda	Leveraging existing systems
9	Development of molecular tests	Uruguay, Vietnam, Singapore, China, South Korea, Germany, USA	Expanding and diversifying diagnostic platforms
10	Development and subsequent use of RDTs	India, Ghana, Bangladesh, Chile, Vietnam, UK, South Korea, China	Expanding and diversifying diagnostic platforms
11	Repurposing manufacturing capacity	USA	Expanding and diversifying diagnostic platforms
12	Training of lab people	Vietnam, Bangladesh, Nigeria, Pakistan, Ethiopia	Human resource enhancement
13	Increased HR base	Bangladesh, DRC, Bhutan, South Africa	Human resource enhancement
14	Legislation to allow for development of tests	USA, South Korea, China, Brazil	Streamlining regulatory & legislative landscape
15	Population wide screening/mass testing		Increasing accessibility

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16	Expanding categories of people to test	Belgium, India, Nigeria, Pakistan, Ghana, Zambia, Bhutan, Iceland, UK, USA	Increasing accessibility
17	Testing only symptomatic and high priority groups	Vietnam, Nigeria, Malawi, France, UK, Brazil	Increasing accessibility
18	Pooled testing	Ghana, Chile	Creative use of technology and innovation
19	Drive through testing/walk in testing	South Korea, Israel, USA	Increasing accessibility
20	Community wide testing using CHWs	South Africa, Mongolia	Increasing accessibility
21	Sample collection and referral of samples	India, DRC, Rwanda	Increasing accessibility
22	Use of veterinary labs	France, Nepal	Leveraging existing systems
23	Massive testing with RDTs	Slovakia, Peru	Expanding and diversifying diagnostic platforms
24	Targeted/hot spot with mass testing	Slovakia, Australia, South Africa	Increasing accessibility
25	Initiatives to support production of diagnostics	USA, South Korea, China	Expanding and diversifying diagnostic platforms
26	Free COVID testing	India, Bhutan, South Korea	Increasing accessibility
27	IT to do surveillance and contact tracing	Ghana, South Korea	Creative use of technology and innovation
29	Drone technology to ship samples	Ghana	Creative use of technology and innovation
20	Continental led efforts (Africa CDC)- pooled purchasing, capacity to country labs, training	Africa, South America, North America by Africa CDC and PAHO	Leveraging regional and continental networks

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	3
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	6,7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	8
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	7
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	NA
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	NA
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	NA
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NA
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	NA
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	9
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	9
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NA
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	9
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	26,27
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	NA
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	NA
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	NA
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	18
	23b	Discuss any limitations of the evidence included in the review.	18
	23c	Discuss any limitations of the review processes used.	18
	23d	Discuss implications of the results for practice, policy, and future research.	18
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Registration started but not completed
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Protocol not prepared
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	19
Competing interests	26	Declare any competing interests of review authors.	19
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	19

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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BMJ Open

A systematic review of strategies adopted to scale up COVID-19 testing in low- middle and high-income countries

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A systematic review of strategies adopted to scale up COVID-19 testing in low, middle and high-income countries

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Number of tables=3
Number of figures=1
Number of supplemental materials=3

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Abstract

Objective: We undertook a systematic review of strategies adopted to scale up COVID-19 testing in countries across income levels in order to identify successful approaches and facilitate learning.

Methods: Scholarly articles in English from PubMed, Google scholar, and google search engine describing strategies used to increase COVID-19 testing in countries were reviewed. Deductive analysis to allocate relevant text from the reviewed publications/reports to the a priori themes was done.

Main results: The review covered 32 countries, including 11 high-income, two upper-middle-income,13 lower-middle-income and six low-income-countries. Most low- and middle-income countries (LMICs) increased the number of laboratories available for testing and deployed sample collection and shipment to the available laboratories. The high-income countries (HICs) i.e., South Korea, Germany, Singapore and USA developed molecular diagnostics with accompanying regulatory and legislative framework adjustments to ensure rapid development and use of the tests. HICs like South Korea leveraged existing manufacturing systems to develop tests, while the LMICs leveraged existing national disease control programmes (HIV, TB, Malaria) to increase testing. Continent-wide, Africa CDC-led collaborations increased testing across most African countries by building capacity through providing testing kits and training.

Conclusion: Strategies taken appear to reflect the existing systems or economies of scale that a particular country could leverage. LMICs, for example, drew upon the infectious disease control programmes already in place to harness expertise and laboratory capacity for COVID-19 testing. There however might have been strategies adopted by other countries but were never published and thus didn't appear anywhere in the searched databases.

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Article summary: Strengths and limitations

- There have been challenges in testing in both developed and developing countries.
- The strength of this systematic review is that it covers many countries across different income categories and provides strategies that worked for countries that other countries could leverage to increase testing now and for future pandemics.
- The findings point to the diverse nature of the laboratory response across both developed and developing nations, mainly driven by the nature of existing capabilities and expertise.
- The major limitation of this systematic review is inability to link the testing strategies to testing rates in each country.

Introduction

At the end of 2019, a disease that presented with pneumonia in the Chinese city of Wuhan emerged.[1] The disease (caused by the SARS-CoV-2 virus) was later named COVID-19 and subsequently declared a global pandemic by the World Health Organisation (WHO) on 11th March 2020. Almost two years into the pandemic, both big and small economies have been hugely affected, and countries will likely continue to suffer for many years after the pandemic's peak. As of 16th December 2021, a total of 273,019,599 cases and 5,338,972 deaths from COVID-19 have been reported.[2] In North America, the United States has notified most cases per 100,000 population while the United Kingdom, India, Brazil, South Africa top the list in Europe, Asia, South America and Africa, respectively.[2]

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Initially, scientists knew little about the disease and the virus, but there have been significant scientific advancements regarding prevention, diagnosis, treatment, and prevention. The world moved from having no or few diagnostic tools to a plethora of diagnostic tools. Furthermore, there has been an unprecedented rapid development and deployment of vaccines, which could be game-changing to ending the pandemic. However, control efforts have been complicated by the emergence of variants, threatening vaccine efficacy.[3]

Beyond the non-pharmacological interventions deployed by countries to combat the pandemic, testing has been deployed to control the spread of COVID-19 by identifying, isolating and treating those infected. Indeed, a few months into the pandemic, WHO called on countries to "test, test and test".[4,5] This was premised on the belief that early testing and contact tracing could interrupt transmission.

Amidst the calls for increased testing, WHO developed an emergency use listing procedure to expedite in vitro diagnostics to support testing,[6] to ensure the availability of diagnostics needed for the public health emergency. WHO also made several recommendations to guide countries on testing. Initial guidance on approaches to testing was based on the number of cases reported per country (sporadic cases, clusters of cases and community transmission). WHO suggested the need for prioritised testing strategies within countries due to the variation in levels of community transmission. [7] With the emergence of new tools like antigen detection rapid diagnostic tests (RDT), the recommendations have been updated. WHO maintained nucleic acid

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amplification tests (NAAT) as the reference standard and recommended antigen RDTs as a complementary strategy.[8]

Despite the advice that testing should underpin the disease control efforts, there have been shortfalls across different countries. There is a noticeable lack of harmonised testing strategies across the globe due to several challenges, including the pre-pandemic laboratory capacity, barriers to access of tests due to increased demand, and challenges with supply. [9] To overcome some of the challenges, countries adopted different strategies to increase the number of tests per capita.

Developing countries (low- and lower-middle-income countries, see results section for definitions) were deemed more likely to bear the heaviest burden of the pandemic, mainly due to fragile health care systems, political instability, economic vulnerability, limited fiscal space, equipment shortages, among others.[10,11] This, however, does not seem to have been the case. For example, the number of cases in most African countries remains reasonably low, with low mortality rates reported.[11,12] While there are many possible reasons why cases have stayed lower than anticipated, it is possible that there is inadequate testing in most of these countries. Indeed, low test per capita rates have been noted in Africa.[13] Some reports have pointed to undercounting of deaths due to COVID-19.[14]

Challenges with testing have also been observed in developed countries,[15,16] indicating that finances alone are not enough to achieve the desired testing capabilities. While the more affluent countries have outcompeted the rest on the COVID-19 diagnostics market,[17] many developing countries have used what they have at their disposal to scale-up testing.

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Future pandemics are always a possibility, and lessons must be learned from the current pandemic to enable robust and expedited responses. There is an opportunity for countries at all income levels to learn lessons from each other. Based on this background, we undertook a systematic review of strategies adopted to scale up COVID-19 testing in countries of all income levels to identify successful approaches and facilitate shared learning.

Methods

The overall aim of the systematic review was to examine strategies adopted to scale up COVID-19 testing in low, middle and high-income-countries and identify drivers underlying the selection of strategy in a given setting.

Inclusion/Exclusion criteria

We included original scholarly articles published in English between December 2019 and September 2021 describing strategies used to increase COVID-19 testing in different countries. Articles included original studies, situational and practice reports, case reports, features, commentaries, viewpoints, reviews, and perspectives. Any scholarly articles that proposed strategies to scale up testing, but with no mention of whether these strategies had been implemented were excluded. The main outcome for the studies was COVID-19 testing strategy and given the diversity in the testing strategies, studies were grouped under themes to support the analysis.

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Search strategy

The search was performed using resources and methods that ensure methodological transparency and reproducibility of the findings, leveraging the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA-P) guidelines.[18] The search strategy was guided by the patient/population, intervention, comparison and outcomes (PICO) approach. For this case, the population was the individuals in the countries, intervention was the COVID-19 testing, while the outcome was COVID-19 testing strategies countries. The study did not have a comparative reference. We asked: *What strategies were adopted to scale up COVID-19 testing in Low- Middle and High-Income countries?* To obtain more relevant articles, we reviewed the bibliograph section of the reviewed articles.

We used the following terms/statements to search the studies for consideration in the systematic review: COVID-19 diagnostics in Asia, South America, Latin America, Europe, North America, COVID-19 diagnostics in high-income countries, COVID-19 diagnostics in middle-income countries, COVID-19 diagnostics in low-income countries, COVID-19 testing in high-income countries, COVID-19 diagnostics in Asia, South America, Latin America, Europe, North America, Increasing COVID-19 testing in Asia, South America, Latin America, Europe, North America, and COVID-19 laboratory testing strategies. The publications returned by the search strategies are summarised in Supplementary Table 1 while the publications included and the corresponding testing strategies are in summarised in Supplementary Table 2.

Data access and management

The investigation team of three researchers accessed the internet sources (PubMed, Google scholar, and google search engine), identified and retrieved all the articles. All

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the retrieved articles were carefully analysed to identify and remove duplicates and irrelevant articles. Researchers through consensus agreed on the final list to consider. The team used the Mendeley referencing manager to store, organise and manage all the articles.

Database/sources searched

The researchers independently searched for scholarly articles (original studies, situational and practice reports, case reports, features, commentaries, viewpoints, reviews and perspectives) in PubMed, Google scholar, and google search engine (all last searched on 17th September 2021). Titles and abstracts were independently screened by researchers based on the inclusion criteria.

Study selection criteria and process

All qualitative or quantitative scholarly articles describing strategies adopted were included in the general screen. Further refinement considered only studies that offered information on the strategy used to increase COVID-19 testing in low, middle and high-income countries were considered further for inclusion.

The selection was carried out by the three researchers working independently. They then convened and reviewed the individual lists and harmonised to agree on one list of studies to be included. The three researchers have postgraduate qualifications in health-related disciplines.

Data extraction

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Both abstracts and full texts were read to identify strategies countries used to increase COVID-19 testing. The outcome was COVID-19 testing strategy and in reviewing the selected articles, researchers sought information on the different approaches that countries used to increase and expand COVID-19 testing. This was achieved through use of a predeveloped codebook that captured the outcome (Supplementary Table 3). The codebook guided the identification of strategies and included themes and narratives of what the researchers were to consider as strategies for increasing testing. Where the researchers found any strategy not captured by the codebook, they noted this. The other variable sought in the reviewed articles was the COVID-19 testing methods used. The findings were later harmonised and summarised in a table that included authors, source of publication, date of publication, country, income category and strategy or strategies used.

Data analysis

Analysis entailed appraising and synthesising texts from the scholarly articles, which were then organised into major themes and content analysis, guided by the predeveloped codebook. We used deductive analysis to allocate relevant text from the reviewed publications and reports to the a priori themes. In the process of allocating text, there were emergent new themes. New and a priori themes included accessibility, diagnostics, regulatory and legislation, integration, personnel, technology and innovation and collaboration.

Patient and Public Involvement

Patients and the public were not involved in any way

Ethics statement

No human participants were included.

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Results

Study selection and characteristics

A total of 174 scholarly articles (134 publications and 40 reports) were identified by the search strategy, of which 59 peer-reviewed publications and 12 reports met the inclusion criteria and were reviewed in detail. The flow diagram (Figure 1) summarises the steps to identify the publications and reports included in the review. Supplementary table 4 represents the details of all papers and reports included and the findings.

Geo-economic coverage of included publications/reports

The selected publications and reports covered six continents (Asia, Africa, North America, Europe, South America, Oceania). There are 32 countries, and of these, 11 are high-income-countries, two are upper-middle-income, 13 are lower-middle-income, and 6 are low-income-countries. Table 1 shows the countries' corresponding Gross Domestic Product (GDP) per capita (USD).[19]

Table 1: Countries included in the systematic review and the corresponding GDP

Country	GDP (USD)
High income countries	
Canada	43241.6
France	38625.1
Belgium	44594.4
Uruguay	15438.4
Slovakia	19156.9
South Korea	31489.1
Australia	51812.2
Chile	13231.7
USA	63543.6
Singapore	59797.8

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Israel	43610.5
Upper middle-income countries	
China	10434.8
South Africa	5090.7
Peru	6126.9
Lower middle-income countries	
India	1900.7
Nepal	1155.1
Nigeria	2097.1
Vietnam	2785.7
Bangladesh	1968.8
Ghana	2328.5
Pakistan	1193.7
Mongolia	4007.3
Zambia	1050.9
Zimbabwe	1128.2
Philippines	3298.8
Bhutan	3122.4
Low-income countries	
Malawi	625.3
DRC	556.8
Rwanda	797.9
Ethiopia	936.3
Sierra Leone	484.5
Uganda	817.0

Testing strategies applied by different countries

Through content analysis, evidence was noted, demonstrating countries adopted diverse strategies to increase testing. These strategies are indicated in table 2 and summarised by income category in table 3, and by individual countries in Supplementary Table 4. Notably, the strategies used were not exclusive, as some countries combined several strategies.

Table 2: Thematic analysis of the different strategies used by the countries

Theme	Strategies
Increasing accessibility	- Increase in no. of labs[9,20–38] - Health technology assessments[20]

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	- Setting up mobile testing units[22,27,39–41] - Population-wide screening/mass testing[37,42,43] - Gradual expansion of categories of people to test[5,22,44] - Testing only symptomatic and high priority groups[26,34,38,41] - Drive through testing/walk-in testing[45–50] - Community-wide testing using CHWs[51–54] - Sample collection and referral of samples[51] - Targeted/hot spot with mass testing[35,55] - Free COVID testing [22,37,43]
Expanding and diversifying diagnostic platforms	- Testing strategy to incorporate use of RDTs [10,20,21,35,56,57] - Development of molecular tests[23,43,58,59] - Development and subsequent use of RDTs [5,20,22,29,43,60,61] - Repurposing manufacturing capacity[62] - Massive testing with RDTs[63] - Initiatives to support production of diagnostics[64,65]
Streamlining regulatory & legislative landscape	- Licensing private providers/labs in testing[5,22,31–33,36,38–40,66] - Allowing research and academic institutions[5,10,25,33,36,38,60,67] - Legislation to allow for development of tests[38,43,62]
Leveraging existing systems	- Leveraging existing platforms for testing diseases [24,25,32,33,36,38,68,69] - Leveraging infrastructure of other disease programs[24,32,36,40,51,70,71] - Leveraging previous systems to develop tests [58,64] - Use of veterinary labs[28,33]
Human resource (HR) enhancement	- Training of laboratory workers [23,24,26,27,30,36,67,68] - Hiring of more personnel [24,37,51]
Creative use of technology & innovation	- Pooled testing[9,35,61] - Information technology (IT) to do surveillance and contact tracing[49,61] - Drone technology to ship samples[40,61,72]
Leveraging regional and continental networks	- Africa CDC, PAHO in training and provision of supplies [11,26,36,73,74]

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Table 3: Strategies by income country

	Countries using the strategy (%)			
Strategy used	**HIC**	**UMIC**	**LMIC**	**LIC**
Increase in number of labs	25.0	66.7	50.0	57.1
Health technology assessments (HTA)	0.0	0.0	8.3	0.0
Testing strategy to incorporate use of RDTs	0.0	33.3	16.7	0.0
Licensing private providers/labs in R&D and testing	37.5	66.7	25.0	14.3
Inclusion of research and academic institutions	31.3	66.7	16.7	28.6
Setting up mobile testing units	6.3	33.3	25.0	0.0
Leveraging existing testing platforms for other diseases	0.0	66.7	41.7	42.9
Development of molecular tests	31.3	0.0	8.3	0.0
Development of RDTs	25.0	0.0	33.3	0.0
Repurposing manufacturing capacity	6.3	0.0	0.0	0.0
Training of lab personnel	0.0	0.0	33.3	14.3
Increased human resource	0.0	33.3	16.7	14.3
Legislation to allow for development of tests	18.8	33.3	0.0	0.0
Leveraging existing infrastructure and systems	6.3	33.3	16.7	42.9
Expanding categories of people to test	18.8	0.0	50.0	0.0
Prioritisation of who to test (symptomatic & high priority groups	12.5	33.3	16.7	14.3
Pooled testing	6.3	0.0	8.3	0.0
Drive through testing/walk in testing	18.8	0.0	0.0	0.0
Community wide testing using CHWs	0.0	33.3	8.3	0.0
Sample collection and referral	0.0	0.0	8.3	28.6
Use of existing veterinary labs	6.3	33.3	0.0	0.0
Massive testing with RDTs	6.3	33.3	0.0	0.0
Identification of hotspots and mass testing	12.5	33.3	0.0	0.0
Initiatives to support production of diagnostics	6.3	0.0	0.0	0.0
Free COVID testing	6.3	0.0	8.3	0.0
IT to do surveillance and contact tracing	6.3	0.0	8.3	0.0
Drone technology to ship samples	0.0	0.0	8.3	0.0
Massive/large scale testing	6.3	0.0	0.0	0.0

HIC=High Income; UMIC=Upper Middle Income; LMIC=Lower middle income; LIC=Low income

The results are presented in the six thematic areas:

(i) Increasing accessibility

Accessibility encapsulates the strategies countries adopted to ensure the testing services had a wider geographical reach closer to the people and that whoever needed a test would be able to get it. Under this theme, most countries increased the number

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of laboratories available for testing, while others increased sample collection points and referral to the available laboratories. The frequency of using these strategies was high in low- and middle-income countries and low in high-income countries. The increment was gradual, with more laboratories being recruited to meet the demand during the pandemic. High-income countries introduced drive through testing to bring testing closer to the people. Sample collection and referral was mainly deployed by the low income and the lower-middle-income countries, while community-wide testing using community health workers was done by the middle-income countries. The high- and upper-middle-income countries increased testing rates by identifying the COVID-19 transmission hot spots in their countries and subsequently undertook mass testing. While the middle- and high-income countries could set up mobile testing units, this wasn't the case for the low-income countries.

(ii) Expanding and diversifying Diagnostic platforms

Under this strategy, the diversity of diagnostics and testing modalities were considered. This included strategies that incorporated the use of rapid diagnostic tests (RDT), development of molecular tests, development and subsequent use of RDTs, repurposing manufacturing capacity to develop tests, community-wide testing with RDT, and initiatives to support the production of diagnostics. Development of molecular diagnostics was undertaken in the high-income countries (South Korea, Germany, Singapore, USA), but not in the low- and middle-income countries except Vietnam, which reported producing a molecular diagnostic to increase testing. In a bid to increase testing, both high-income (South Korea, Chile, United Kingdom) and lower-middle-income countries (India, Bangladesh, Vietnam, Ghana) developed RDTs. Adopting rapid diagnostic tests increased testing rates in these countries. Only one country (USA) reported repurposing manufacturing capacity to manufacture testing

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kits. Both high- and upper-middle-income countries carried out massive testing with RDTs.

(iii) Streamlining regulatory and the legislative landscape

Streamlining the regulatory and legislative landscape evolved in two ways: Accelerating the manufacturing process and increasing the number of laboratories available for testing. Governments of high-income countries adjusted the regulatory and legislative frameworks to ensure rapid development and the use of the developed tests. For example, the USA and South Korea deployed emergency use authorisation and saved the manufacturers from bureaucratic and administrative inefficiencies associated with the approval process for the diagnostics. The governments also linked test manufacturers with the clinical bodies to help fast track the validation process. As a result, tests were readily available for use in a short time frame. In Singapore, public-private partnerships were instrumental in ensuring the massive production of tests.

Governments also made regulatory and legislative adjustments to ensure more laboratories became available for COVID-19 testing. The added laboratories included the private, research and academic laboratories. This was noted mainly in the upper-middle, lower-middle- and low-income countries. For example, France, brought private providers later in the pandemic to avoid possibility of having non-harmonised reporting.

Also, as part of the regulations and strategies to attract private test developers, the high-income countries (USA, South Korea) deployed initiatives to interest private developed in manufacturing. Such initiatives included easing the approval processes and assurance that the governments would buy the developed kits.

(iv) Leveraging existing systems

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The countries leveraged existing systems in two ways. One, using the current systems and frameworks to scale up development of diagnostics and two, using the existing systems to incorporate COVID-19 testing.

High-income countries like South Korea leveraged structures established during the previous epidemics like MERS to be able to develop tests in a short time. Low- and middle-income- countries integrated COVID-19 testing within the existing disease systems to increase testing. The countries leveraged the testing systems available for other diseases like Tuberculosis, HIV and Influenza to embark on testing for COVID-19. In most of these countries, there were existing platforms for HIV viral load monitoring and Xpert MTB/RIF testing, which were leveraged to test for COVID-19. The integration also leveraged already set up systems to increase testing. For example, Nigeria leveraged the established polio systems to ensure the collection and transport of samples to the testing facilities.[75]

(v) Human resource enhancement

Countries undertook human resource initiatives. Such initiatives included recruiting more health workers and increasing their capacity through training. Recruitment of more personnel happened mainly in the upper-middle-income countries, followed by lower-middle-income countries and then the low-income countries. The trend was slightly different when it came to increasing capacity through training. The training was conducted mainly in lower-middle- and low-income countries.

(vi) Creative use of technology and innovation

Countries such as South Korea and Ghana leveraged technology to increase COVID-19 testing. South Korea was the first country to roll out a contact tracing and surveillance application to help increase testing. Ghana reported using a contact tracing application to increase testing. Ghana also increased testing by using drones

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to transport samples from the field to the testing facilities established. Furthermore, Ghana innovatively used pooled testing to increase the number of samples tested with limited kits. Only India reported undertaking health technology assessment (HTA) on a locally developed COVID-19 diagnostic, whereby they assessed the impact of a rapid diagnostic test on health systems in dimensions of efficiency, equity and quality guided by the HTA framework.[20]

(vii) Leveraging regional and continental networks

Whereas most of the interventions happened at the country level, some of the efforts resulted from continent-wide collaborations led by the continental bodies. By far, the most extensive continental-led testing response was on the African continent, where the Africa CDC increased testing across most African countries by building capacity through the provision of testing kits and training. In addition, through the Africa Task Force for Corona Virus (AFCOR) and the Partnership to Accelerate COVID-19 testing (PACT), the Africa CDC helped increase testing. The Pan African Health Organisation (PAHO) led the efforts in South America and North America by ordering and providing test kits, conducting training, and leveraging the national influenza centres to increase the laboratory network.

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Discussion

This systematic review of strategies adopted to scale up COVID-19 testing has revealed strategies unique to some and common to all countries. We note that the choice of strategy was mainly driven by the previous outbreak management system to leverage the country's income and manufacturing capacity. The adopted strategies fall under six thematic areas: increasing accessibility, expanding, and diversifying

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diagnostic platforms, streamlining regulatory and legislative landscape, leveraging existing systems, human resource enhancement, creative use of technology and innovation and leveraging regional and continental networks.

The COVID-19 pandemic caught many countries unawares, irrespective of their income status. For example, in the USA and the UK, initial testing responses are marked by the absence of a clear testing strategy, defective tests, and sluggish inclusion of private laboratories.[59,66] These obstacles were later overcome, private and academic institutions were included and the categories of people prioritised for testing was expanded.[66]

Countries in a bid to avail tests to their people, took on strategies to improve access. Most countries gradually increased the number of testing laboratories. This was largely done in the low- and middle-income countries and to a lesser extent in high-income countries. The high-income countries introduced drive-through/walk-in testing. The lower- and upper-middle-income countries undertook community wide testing using community health care workers, a strategy supported by the decentralised nature of the health care system in most low- and middle-income countries. Countries like Zambia, South Africa used community health workers with good results.[44]

In a bid to save resources, countries adopted the symptomatologic approach to ration testing to only those exhibiting COVID-19 like symptoms. While this may have saved resources, it most likely missed many asymptomatic cases that are now thought to have driven the transmission of COVID-19. Both high-, middle- and low-income countries applied the approach, and gradually expanded the categories of individuals to test. HICs expanded testing faster than LMICs though it is important to note that apart from contact tracing, test seeking is still largely driven by experience of symptoms. Despite health care workers being at the frontline and the need to have them tested, the United Kingdom did not prioritise them until later in the pandemic.[76] Whereas large economies like South Korea and Singapore opened up testing to everyone and did mass testing, several African, Asian and European countries across all income categories limited testing to the symptomatic and the high priority groups, and only expanded the categories later. This could largely have been due to the

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logistical challenges and the need to use the supplies available sparingly as it was hard for most of them to compete.[22,35,37,44,77]

Countries sought to expand and diversify the diagnostic platforms available for COVID-19 testing largely through the development and/or the adoption of molecular and rapid diagnostic tests. For the high-income countries, this was achieved through the development of molecular and rapid diagnostic tests, while for other countries, it was mainly through rapid diagnostic tests. All the molecular (RT-PCR) diagnostics were developed in the high-income countries[58,59,62,65,66] and this to some extent protected them from shortages of RT-PCR testing supplies that other countries had to contend with. RT-PCR testing required highly specialised laboratories that were few in most low- and middle-income countries. Some of these countries adopted rapid diagnostic tests (RDTs) in their testing algorithms as an interim measure. These were either imported or made from within. Some lower- and middle-income countries developed RDTs, and these helped increase testing.[20,61] and were gradually introduced in other low-income countries to boost testing.[9,21,35]

Some countries enhanced their regulatory and legislative landscape to maximise the number of tests within the shortest timeframe. HICs led the introduction of legislation mainly to increase manufacturing capacity, adoption of new diagnostic tests and accreditation of non-traditional facilities for involvement in testing COVID-19. In the USA, legislation to modify the regulatory framework governing laboratory developed tests led to more tests being introduced later.[62] However, this came several months into the pandemic. The federal government was blamed for the bureaucracy that prevented the provision of tests by academic and private laboratories, and product developers.[78] In an emergency, an early response is key, and lifting non-critical administrative barriers without compromising oversight can help expedite the delivery of vital products like diagnostics.[79] As part of regulation and licencing, countries included private, academic and research laboratories in the response.

There were differences in the engagement of private stakeholders between high-income countries and the low- and middle-income countries and the point at which they were engaged. While the high-income countries private sector was engaged in the early days of the pandemic as test developers [65,66,80], in the low- and middle-

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income countries, they were engaged in testing.[31–33,39,40,81] In most of the high-income countries the private laboratories were engaged in testing quite late.[66,77] Countries called on academic institutions to support the testing efforts. This was noted across all countries i.e., low-, middle- and high-income countries.[29,60,67] The academic institutions in most countries have molecular platforms they use mainly for research purposes, and when leveraged could help supplement government level efforts to increase testing. Furthermore, most academic institutions in low- and middle-income countries have collaborative links with partners in the developed world. Such networks could have been leveraged to ensure the availability of kits for testing. Early in the pandemic, the United Kingdom engaged the academic institutions by asking them to avail their platforms for use in public facilities. This was later changed and instead the academic institutions were called on to do the testing.[5]

Countries also took advantage of available systems and previous experiences to mount responses to increase COVID-19 testing. Whereas the high-income countries used prior experience to develop tests, the low-and middle-income countries leveraged the systems available to increase testing. The existing systems worked in favour of low- and middle-income-countries as they could tap into these to effect testing. The Polymerase chain reaction (PCR) testing footprint within the disease programs like TB, HIV, Influenza, and Schistosomiasis were used by most of the low- and middle-income countries.[33,36,51,68] All they needed were supplies and the collaborations and donations helped with this.[11,36,82] With most of the low and middle income countries accounting for the highest burden of TB[83], there have been progressive investments in diagnostics for TB and these have been leveraged for COVID-19 testing. Korea was able to rapidly develop testing kits by leveraging the test development strategy used for MERS.[84]

Human resources underpin most of the testing needed by the countries, as having equipment alone is not enough. As a result, countries recruited and trained more personnel.[11,24,30,51,68] Recruitment of extra personnel was mainly reported by the upper middle-income countries followed by the lower middle-income countries, while the training of personnel was reported mainly by the lower middle-income countries followed by the low-income countries. Human resource challenges in low- and middle-income countries have previously been documented[85,] and the use of this as a

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strategy was possibly to meet the short falls in number available to run the COVID-19 tests. The recruitment and training of personnel was not noted in any of the high-income countries, possibly an indicator that these countries are not facing challenges of limited human resources for health.

Some countries were creative and innovative in their approach. In order to save the limited logistics, Ghana adopted pooled testing.[61] Specimen pooling is a method of screening a large number of patients for an infection and typically involves combining multiple patient specimens into a single test tube than testing multiple samples in separate reaction tubes.[86] This approach enabled Ghana to increase the number of tests done to the extent that in Africa they posted the highest number of tests done only behind South Africa.[61]

There were strategies at continental level, and these were overseen by the relevant bodies. By far, the greatest demonstration of continent-led efforts was in Africa where the African Centres for Disease Control and Prevention (Africa CDC). Africa CDC demonstrated the need for harmonisation, solidarity, and coordination at a continental level. They launched a joint continental strategy, the partnership to accelerate COVID-19 testing, purchased kits and equipped several African laboratories.[11,82] African countries were outcompeted on the global supplies market and through the Africa CDC, pooled purchasing was implemented so that African countries competed as a block (as one large customer).[36] Similar efforts are noted in South America where the Pan American Health Organisation (PAHO) oversaw country level coordination, planning and monitoring. It mobilised experts and furnished countries with laboratory supplies for detection of the virus.[87]

The review has revealed that while the response was reactive, mostly trial and error in the early days of the COVID-19 pandemic, it yielded strategies that could be leveraged to support implementation of evidence-based practices in the present and future epidemic-pandemic scenarios. We have noted how countries drew upon the existing systems such as standard operating procedures (SOPs) used in responding to previous epidemics and/or refocusing manufacturing base to produce testing kits. The lesson that can be learned from this is that preparedness is crucial to put systems in place that can be drawn upon to effectively respond to disease outbreaks and

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epidemics. Testing was only part of the response that included clinical management of patients, contact tracing, quarantining, lockdowns to mention a few. This implies that preparedness should be holistic including heath system strengthening (laboratory, clinical, referral system, disease surveillance, procurement and governance), socioeconomic empowerment of people and a flexible legislative and regulatory framework to enable rapid public health intervention in periods of health emergency.

The limitations of this review are that there might have been strategies adopted by other countries but were never published and thus didn't appear anywhere in the searched databases. Further, the review only considered scholarly articles in English, and thus there might be underrepresentation of strategies used in the non-English speaking countries.

Conclusion

All countries regardless of their income level, had unique challenges associated with scale-up of COVID-19 testing, and the review has revealed evidence that countries used different strategies to increase testing. Whereas the low- and middle-income countries struggled to increase the testing rates as reflected by the low testing rates, they were able to post optimal testing rates because of experience in dealing with disease outbreaks and leveraging of the existing disease control programs. The high-income countries in Europe and North America, despite their economic advantage, also struggled having enough tests available in the short to medium term of COVID-19 pandemic. This was not the case in Asia where the high-income economies leveraged the previous disease outbreak experience like SARS, MERS to have in place diagnostics in a short period of time and quickly put in place effective testing strategies. Country's epidemic handling experience, and manufacturing and financial landscape were critical in informing strategy to scale up testing. Learning lessons from the COVID-19 pandemic and strengthening health systems will equip the world to respond more effectively to any future pandemics.

CONTRIBUTORSHIP

WM, WS, VS: Conceptualisation, funding acquisition, project administration, data curation, formal analysis; WM, WS, BO, VS, AR, BK, PT, BM: Investigation,

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methodology, manuscript revision; WM: data curation, formal analysis, original draft writing

COMPETING INTERESTS

None declared

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DATA SHARING

Data are available upon reasonable request. will be made available to others upon reasonable request. A formal request should be sent via email to the corresponding author Dr Winters Muttamba at wm55@st-andrews.ac.uk

Additional information

This study was not registered and a full review protocol was not written as it was originally conducted as a rapid review and later changed to a full systematic review.

Figure 1: Flow diagram of included publications

References

1. Shi Y, Wang G, Cai X peng, et al. An overview of COVID-19. *Journal of Zhejiang University-SCIENCE B 2020 21:5*. 2020;21(5):343-360. doi:10.1631/JZUS.B2000083
2. COVID-19 Map - Johns Hopkins Coronavirus Resource Center. Accessed December 17, 2021. https://coronavirus.jhu.edu/map.html

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

3. Tregoning JS, Flight KE, Higham SL, Wang Z, Pierce BF. Progress of the COVID-19 vaccine effort: viruses, vaccines and variants versus efficacy, effectiveness and escape. *Nature Reviews Immunology*. Published online 2021. doi:10.1038/S41577-021-00592-1
4. WHO Director-General's opening remarks at the media briefing on COVID-19 - 16 March 2020. Accessed September 28, 2021. https://www.who.int/director-general/speeches/detail/who-director-general-s-opening-remarks-at-the-media-briefing-on-covid-19---16-march-2020
5. Rough E. Coronavirus: Testing for Covid-19. *Number CBP*. 2020;8897. Accessed September 27, 2021. www.parliament.uk/commons-library|intranet.parliament.uk/commons-library|papers@parliament.uk|@commonslibrary
6. Coronavirus disease (COVID-19) Pandemic — Emergency Use Listing Procedure (EUL) open for IVDs | WHO - Prequalification of Medical Products (IVDs, Medicines, Vaccines and Immunization Devices, Vector Control). Accessed October 20, 2021. https://extranet.who.int/pqweb/vitro-diagnostics/coronavirus-disease-covid-19-pandemic-%E2%80%94-emergency-use-listing-procedure-eul-open
7. Laboratory testing strategy recommendations for COVID-19: interim guidance, 21 March 2020. Accessed August 5, 2022. https://apps.who.int/iris/handle/10665/331509
8. Recommendations for national SARS-CoV-2 testing strategies and diagnostic capacities Interim guidance. Published online 2021.
9. Benítez MA, Velasco C, Sequeira AR, Henríquez J, Menezes FM, Paolucci F. Responses to COVID-19 in five Latin American countries. *Health Policy and Technology*. 2020;9(4):525-559. doi:10.1016/J.HLPT.2020.08.014
10. Garcia PJ, Alarcón A, Bayer A, et al. COVID-19 Response in Latin America. *The American Journal of Tropical Medicine and Hygiene*. 2020;103(5):1765. doi:10.4269/AJTMH.20-0765
11. Medinilla A, Byiers B, Apiko P. ecdpm's African regional responses to COVID-19. Published online 2020. Accessed September 20, 2021. www.ecdpm.org/dp272
12. Total confirmed COVID-19 cases. Accessed September 28, 2021. https://ourworldindata.org/grapher/covid-cases-income
13. Hasell J, Mathieu E, Beltekian D, et al. A cross-country database of COVID-19 testing. *Scientific Data*. 2020;7(1). doi:10.1038/S41597-020-00688-8
14. A Continent Where the Dead Are Not Counted - The New York Times. Accessed November 29, 2021. https://www.nytimes.com/2021/01/02/world/africa/africa-coronavirus-deaths-underreporting.html
15. FDA Emergency Use Authorization (EUA) Guidance for COVID-19 Testing. Accessed September 28, 2021. https://crsreports.congress.gov
16. Coronavirus (COVID-19): scaling up our testing programmes - GOV.UK. Accessed August 5, 2022. https://www.gov.uk/government/publications/coronavirus-covid-19-scaling-up-testing-programmes/coronavirus-covid-19-scaling-up-our-testing-programmes
17. Nkengasong J. Let Africa into the market for COVID-19 diagnostics. *Nature*. 2020;580(7805):565-565. doi:10.1038/D41586-020-01265-0
18. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *The BMJ*. 2021;372. doi:10.1136/BMJ.N71
19. GDP per capita (current US$) | Data. Accessed October 20, 2021. https://data.worldbank.org/indicator/NY.GDP.PCAP.CD
20. Mukherjee K. Integrating technology, innovation and policy: COVID-19 and HTA. *Health Policy and Technology*. 2021;10(1):16-20. doi:10.1016/J.HLPT.2021.01.003

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

21. Budhathoki P, Shrestha DB, Khadka S, Giri S. COVID-19 Status in Nepal and the Way Forward. *Europasian Journal of Medical Sciences* . 2020;2(2):99-104. doi:10.46405/EJMS.V2I2.75
22. Group GC 19 S. Combating the COVID-19 pandemic in a resource-constrained setting: insights from initial response in India. *BMJ Global Health*. 2020;5(11):e003416. doi:10.1136/BMJGH-2020-003416
23. Nguyen TV, Tran QD, Phan LT, et al. In the interest of public safety: rapid response to the COVID-19 epidemic in Vietnam. *BMJ Global Health*. 2021;6(1):e004100. doi:10.1136/BMJGH-2020-004100
24. Saha S, Pai M. Can COVID-19 innovations and systems help low- and middle-income countries to re-imagine healthcare delivery? *Med*. 2021;2(4):369-373. doi:10.1016/J.MEDJ.2021.02.008
25. Munharo S, Nayupe SF, Patel P. Challenges of COVID-19 testing in low-middle income countries (LMICs) Challenges of COVID-19 testing in low-middle income countries (LMICs): the case of Malawi View project Patrick Mbulaje Centre for the Development of People (CEDEP) 4 PUBLICATIONS 3 CITATIONS SEE PROFILE. doi:10.21037/jlpm-20-84
26. Amzat J, Aminu K, Kolo VI, Akinyele AA, Ogundairo JA, Danjibo MC. Coronavirus outbreak in Nigeria: Burden and socio-medical response during the first 100 days. *International Journal of Infectious Diseases*. 2020;98:218-224. doi:10.1016/J.IJID.2020.06.067
27. Waris A, Atta UK, Ali M, Asmat A, Baset A. COVID-19 outbreak: current scenario of Pakistan. *New Microbes and New Infections*. 2020;35:100681. doi:10.1016/J.NMNI.2020.100681
28. Dhakal S, Karki S. Early Epidemiological Features of COVID-19 in Nepal and Public Health Response. *Frontiers in Medicine*. 2020;0:524. doi:10.3389/FMED.2020.00524
29. Anwar S, Nasrullah M, Hosen MJ. COVID-19 and Bangladesh: Challenges and How to Address Them. *Frontiers in Public Health*. 2020;0:154. doi:10.3389/FPUBH.2020.00154
30. Nawaz A, Su X, Barkat MQ, et al. Epidemic Spread and Its Management Through Governance and Leadership Response Influencing the Arising Challenges Around COVID-19 in Pakistan—A Lesson Learnt for Low Income Countries With Limited Resource. *Frontiers in Public Health*. 2020;8. doi:10.3389/FPUBH.2020.573431/PDF
31. Siddiqui AF, Wiederkehr M, Rozanova L, Flahault A. Situation of India in the COVID-19 Pandemic: India's Initial Pandemic Experience. *International Journal of Environmental Research and Public Health 2020, Vol 17, Page 8994*. 2020;17(23):8994. doi:10.3390/IJERPH17238994
32. Al-Mustapha AI, Tijani AA, Oyewo M, et al. Nigeria's race to zero COVID-19 cases: True disease burden or testing failure? *Journal of Global Health*. 2021;11. doi:10.7189/JOGH.11.03094
33. Acheampong G, Owusu M, Nkrumah B, et al. Laboratory capacity in COVID-19 diagnosis and the need to enhance molecular testing in Ghana. *https://doi.org/101080/237794972021908157*. 2021;6(1):10-17. doi:10.1080/23779497.2021.1908157
34. Ferree KE, Harris AS, Dulani B, Kao K, Lust E, Metheney E. Stigma, Trust, and procedural integrity: Covid-19 testing in Malawi. *World Development*. 2021;141:105351. doi:10.1016/J.WORLDDEV.2020.105351
35. Nyazika TK, Kaela R, Mugoni M, et al. Implementation of Antibody Rapid Diagnostic Testing versus Real-Time Reverse Transcription-PCR Sample Pooling in the

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Screening of COVID-19: a Case of Different Testing Strategies in Africa. *mSphere*. 2020;5(4). doi:10.1128/MSPHERE.00524-20
36. Ondoa P, Kebede Y, Loembe MM, et al. COVID-19 testing in Africa: lessons learnt. *The Lancet Microbe*. 2020;1(3):e103-e104. doi:10.1016/S2666-5247(20)30068-9
37. Dorji T. The Gross National Happiness Framework and the Health System Response to the COVID-19 Pandemic in Bhutan. *The American Journal of Tropical Medicine and Hygiene*. 2021;104(2):441. doi:10.4269/AJTMH.20-1416
38. K K, MM B, R C, et al. Testing COVID-19 in Brazil: fragmented efforts and challenges to expand diagnostic capacity at the Brazilian Unified National Health System. *Cadernos de saude publica*. 2021;37(3). doi:10.1590/0102-311X00277420
39. Ogbolosingha AJ, Singh A. COVID-19 pandemic: Review of impediments to public health measures in Sub-Saharan Africa. Published online 2020. doi:10.5455/ajpmph.20200512020014
40. Adepoju P. Africa's struggle with inadequate COVID-19 testing. *The Lancet Microbe*. 2020;1(1):e12. doi:10.1016/S2666-5247(20)30014-8
41. How resilient have European health systems been to the COVID-19 crisis? Published online November 19, 2020. doi:10.1787/85E4B6A1-EN
42. Mahase E. Covid-19: Mass testing in Slovakia may have helped cut infections. *BMJ*. 2020;371:m4761. doi:10.1136/BMJ.M4761
43. Duchâtel M. ABOUT THE AUTHORS. Published online 2015. Accessed September 27, 2021. www.institutmontaigne.org/en
44. Chipimo PJ, Barradas DT, Kayeyi N, et al. First 100 Persons with COVID-19 — Zambia, March 18–April 28, 2020. *Morbidity and Mortality Weekly Report*. 2020;69(42):1547. doi:10.15585/MMWR.MM6942A5
45. Choi S, Han C, Lee J, Kim SI, Kim IB. Innovative screening tests for COVID-19 in South Korea. *Clinical and Experimental Emergency Medicine*. 2020;7(2):73. doi:10.15441/CEEM.20.032
46. Kwon KT, Ko JH, Shin H, Sung M, Kim JY. Drive-Through Screening Center for COVID-19: a Safe and Efficient Screening System against Massive Community Outbreak. *Journal of Korean Medical Science*. 2020;35(11):123. doi:10.3346/JKMS.2020.35.E123
47. Kim E. Drawing on Israel's Experience Organizing Volunteers to Operationalize Drive-Through Coronavirus Testing Centers. *Disaster Medicine and Public Health Preparedness*. 2020;14(6):812-814. doi:10.1017/DMP.2020.104
48. Evans LK, Shinagawa A, Sutton S, Calvo L. COVID-19 Drive-Through Point of Screening and Testing (POST) System: A Safe, Efficient, and Adaptable Model for Nasopharyngeal Swab Collection. *Disaster Medicine and Public Health Preparedness*. Published online 2020:1-7. doi:10.1017/DMP.2020.313
49. Madhi SA, Baillie V, Cutland CL, et al. Safety and efficacy of the ChAdOx1 nCoV-19 (AZD1222) Covid-19 vaccine against the B.1.351 variant in South Africa. *Alex Sigal*. 2021;13:2. doi:10.1101/2021.02.10.21251247
50. Shah A, Challener D, Tande AJ, et al. Drive-Through Testing: A Unique, Efficient Method of Collecting Large Volume of Specimens During the SARS-CoV-2 (COVID-19) Pandemic. *Mayo Clinic Proceedings*. 2020;95(7):1420-1425. doi:10.1016/J.MAYOCP.2020.04.030
51. Nachega JB, Grimwood A, Mahomed H, et al. From Easing Lockdowns to Scaling Up Community-based Coronavirus Disease 2019 Screening, Testing, and Contact Tracing in Africa—Shared Approaches, Innovations, and Challenges to Minimize Morbidity and Mortality. *Clinical Infectious Diseases*. 2021;72(2):327-331. doi:10.1093/CID/CIAA695

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

52. Odendaal N. Constructing an "infrastructure of care" – understanding the institutional remnants and socio-technical practices that constitute South Africa's Covid-19 response. *https://doi.org/101080/0272363820201807166*. 2020;42(3):391-398. doi:10.1080/02723638.2020.1807166
53. Massinga Loembé M, Tshangela A, Salyer SJ, Varma JK, Ouma AEO, Nkengasong JN. COVID-19 in Africa: the spread and response. *Nature Medicine 2020 26:7*. 2020;26(7):999-1003. doi:10.1038/s41591-020-0961-x
54. Mendelson M, Madhi S. South Africa's coronavirus testing strategy is broken and not fit for purpose: It's time for a change. *South African Medical Journal*. 2020;110(6):429-431. doi:10.7196/SAMJ.2020.V110I6.14902
55. Boretti A. The Increasing Number of Infected in Victoria, Australia since June 15, 2020, is the Result of Over-Testing and Over-Controlling without Safety. *Integrative Journal of Medical Sciences*. 2020;7:187. doi:10.15342/IJMS.7.187
56. Herrera-Añazco P, Uyen-Cateriano A, Mezones-Holguin E, et al. Some lessons that Peru did not learn before the second wave of COVID-19. *The International Journal of Health Planning and Management*. 2021;36(3):995-998. doi:10.1002/HPM.3135
57. Laxminarayan R, Jameel S, Sarkar S. India's Battle against COVID-19: Progress and Challenges. *The American Journal of Tropical Medicine and Hygiene*. 2020;103(4):1343. doi:10.4269/AJTMH.20-0992
58. Kuguyo O, Kengne AP, Dandara C. Singapore COVID-19 Pandemic Response as a Successful Model Framework for Low-Resource Health Care Settings in Africa? *https://home.liebertpub.com/omi*. 2020;24(8):470-478. doi:10.1089/OMI.2020.0077
59. Jamison DT, Wu KB. The East–West Divide in Response to COVID-19. *Engineering*. Published online June 12, 2021. doi:10.1016/J.ENG.2021.05.008
60. Taylor L. Uruguay is winning against covid-19. This is how. *BMJ*. 2020;370. doi:10.1136/BMJ.M3575
61. Sibiri H, Zankawah SM, Prah D. Coronavirus diseases 2019 (COVID-19) response: Highlights of Ghana's scientific and technological innovativeness and breakthroughs. *Ethics, Medicine and Public Health*. 2020;14:100537. doi:10.1016/J.JEMEP.2020.100537
62. Vandenberg O, Martiny D, Rochas O, van Belkum A, Kozlakidis Z. Considerations for diagnostic COVID-19 tests. *Nature Reviews Microbiology 2020 19:3*. 2020;19(3):171-183. doi:10.1038/s41579-020-00461-z
63. Frnda J, Durica M. On Pilot Massive COVID-19 Testing by Antigen Tests in Europe. Case Study: Slovakia. *Infectious Disease Reports 2021, Vol 13, Pages 45-57*. 2021;13(1):45-57. doi:10.3390/IDR13010007
64. Vandenberg O, Martiny D, Rochas O, van Belkum A, Kozlakidis Z. Considerations for diagnostic COVID-19 tests. *Nature Reviews Microbiology 2020 19:3*. 2020;19(3):171-183. doi:10.1038/s41579-020-00461-z
65. Tromberg BJ, Schwetz TA, Pérez-Stable EJ, et al. Rapid Scaling Up of Covid-19 Diagnostic Testing in the United States — The NIH RADx Initiative. *https://doi.org/101056/NEJMsr2022263*. 2020;383(11):1071-1077. doi:10.1056/NEJMSR2022263
66. Alexander M, Unruh L, Koval A, Belanger W. United States response to the COVID-19 pandemic, January–November 2020. *Health Economics, Policy and Law*. Published online 2021:1-14. doi:10.1017/S1744133121000116
67. Assefa N, Hassen JY, Admassu D, et al. COVID-19 Testing Experience in a Resource-Limited Setting: The Use of Existing Facilities in Public Health Emergency Management. *Frontiers in Public Health*. 2021;0:718. doi:10.3389/FPUBH.2021.675553

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

68. Abera A, Belay H, Zewude A, et al. Establishment of COVID-19 testing laboratory in resource-limited settings: challenges and prospects reported from Ethiopia. *https://doi.org/101080/1654971620201841963*. 2020;13(1). doi:10.1080/16549716.2020.1841963
69. Lau LL, Hung N, Wilson K. COVID-19 response strategies: considering inequalities between and within countries. *International Journal for Equity in Health 2020 19:1*. 2020;19(1):1-3. doi:10.1186/S12939-020-01254-9
70. Erkhembayar R, Dickinson E, Badarch D, et al. Early policy actions and emergency response to the COVID-19 pandemic in Mongolia: experiences and challenges. *The Lancet Global Health*. 2020;8(9):e1234-e1241. doi:10.1016/S2214-109X(20)30295-3
71. Dzinamarira T, Dzobo M, Chitungo I. COVID-19: A perspective on Africa's capacity and response. *Journal of Medical Virology*. 2020;92(11):2465-2472. doi:10.1002/JMV.26159
72. Sarfo AK, Karuppannan S. Application of Geospatial Technologies in the COVID-19 Fight of Ghana. *Transactions of the Indian National Academy of Engineering 2020 5:2*. 2020;5(2):193-204. doi:10.1007/S41403-020-00145-3
73. Kavanagh MM, Erondu NA, Tomori O, et al. Access to lifesaving medical resources for African countries: COVID-19 testing and response, ethics, and politics. *The Lancet*. 2020;395(10238):1735-1738. doi:10.1016/S0140-6736(20)31093-X
74. Organization PAH. Pan American Health Organization Response to COVID-19 in the Americas, 17 January to 31 May 2020. Published online July 6, 2020. Accessed September 27, 2021. https://iris.paho.org/handle/10665.2/52453
75. COVID-19 shows value of polio infrastructure to support resilient health systems – GPEI. Accessed August 5, 2022. https://polioeradication.org/news-post/covid-19-shows-value-of-polio-infrastructure-to-support-resilient-health-systems/
76. Coronavirus (COVID-19): scaling up our testing programmes - GOV.UK. Accessed August 5, 2022. https://www.gov.uk/government/publications/coronavirus-covid-19-scaling-up-testing-programmes/coronavirus-covid-19-scaling-up-our-testing-programmes
77. Desson Z, Weller E, McMeekin P, Ammi M. An analysis of the policy responses to the COVID-19 pandemic in France, Belgium, and Canada. *Health Policy and Technology*. 2020;9(4):430-446. doi:10.1016/J.HLPT.2020.09.002
78. How the government delayed coronavirus testing - CNNPolitics. Accessed October 22, 2021. https://edition.cnn.com/2020/04/09/politics/coronavirus-testing-cdc-fda-red-tape-invs/index.html
79. Note by the Secretariat in consultation with the Chairs and the Bureaus of the Regulatory Policy Committee and the Network of Economic Regulators Regulatory Quality and COVID-19: Managing the Risks and Supporting the Recovery. Published online 2020.
80. Lee D, Lee J. Testing on the move: South Korea's rapid response to the COVID-19 pandemic. *Transportation Research Interdisciplinary Perspectives*. 2020;5:100111. doi:10.1016/J.TRIP.2020.100111
81. Kebede A, Lanyero B, Beyene B, et al. Expanding molecular diagnostic capacity for COVID-19 in Ethiopia: operational implications, challenges and lessons learnt. *The Pan African Medical Journal*. 2021;38:1-11. doi:10.11604/PAMJ.2021.38.68.27501
82. Patterson AS, Balogun E. African Responses to COVID-19: The Reckoning of Agency? *African Studies Review*. 2021;64(1):144-167. doi:10.1017/ASR.2020.122
83. Global Tuberculosis Report 2021. Accessed October 22, 2021. https://www.who.int/teams/global-tuberculosis-programme/tb-reports/global-tuberculosis-report-2021

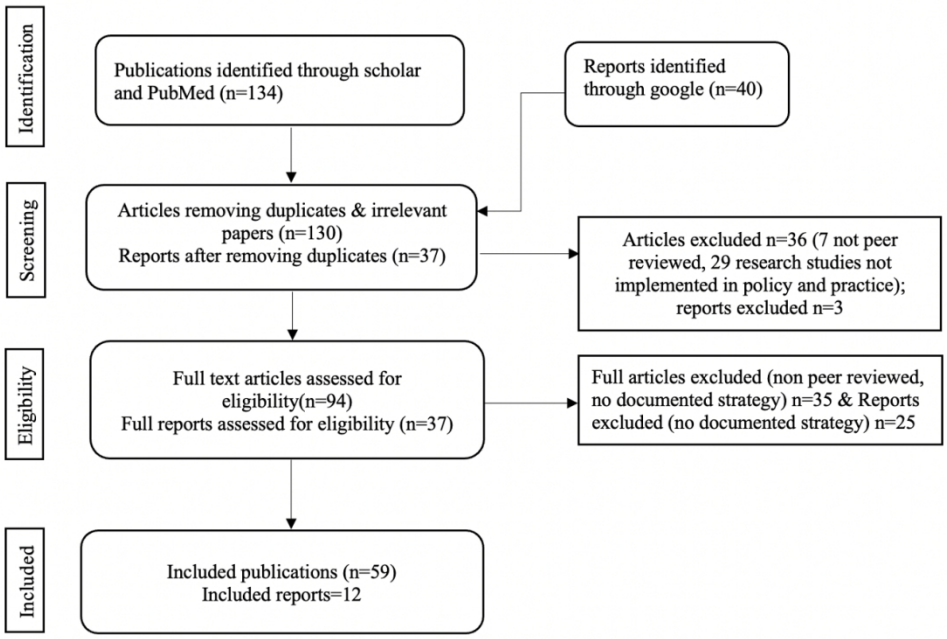
For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

84. South Korea's Response to COVID-19 | FDA. Accessed October 22, 2021. https://www.fda.gov/medical-devices/coronavirus-covid-19-and-medical-devices/south-koreas-response-covid-19
85. Hongoro C, McPake B. How to bridge the gap in human resources for health. *The Lancet*. 2004;364(9443):1451-1456. doi:10.1016/S0140-6736(04)17229-2
86. Wacharapluesadee S, Kaewpom T, Ampoot W, et al. Evaluating the efficiency of specimen pooling for PCR-based detection of COVID-19. *Journal of Medical Virology*. 2020;92(10):2193-2199. doi:10.1002/JMV.26005
87. Pan American Health Organization Response to COVID-19 in the Americas, 17 January to 31 May 2020 - PAHO/WHO | Pan American Health Organization. Accessed September 27, 2021. https://www.paho.org/en/documents/pan-american-health-organization-response-covid-19-americas-17-january-31-may-2020

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Figure 1: Flow diagram of included Publications



301x228mm (118 x 118 DPI)

Supplementary Table 1: Publications from the search strategy

Search strategy	Number of publications
COVID-19 diagnostics in Asia	42
COVID-19 diagnostics South America	26
COVID-19 diagnostics Latin America	17
COVID-19 diagnostics Europe	29
COVID-19 diagnostics North America	30
COVID-19 diagnostics in middle-income countries	37
COVID-19 diagnostics in low-income countries	45
COVID-19 testing in high-income countries	34
Increasing COVID-19 testing in Asia	28
Increasing COVID-19 testing in South America	11
Increasing COVID-19 testing in Latin America	5
Increasing COVID-19 testing in Europe	16
Increasing COVID-19 testing in North America	17
COVID-19 laboratory testing strategies	56

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A systematic review of COVID-19 diagnostics uptake pathways in Low- Middle Income and High-Income countries							
Author(s)	Title of the article	Journal name	Type of article	Month & Year Published	Country	Income category (LMIC or HIC)	How COVID-19 testing was increased
Kanchan Mukherjee	Integrating technology, innovation and policy: COVID-19 and HTA	Elesevier- Health policy and technologuy	research article	Feb-21	India	LMIC	1)Department of science and technology launched initiatives to support COVID-19 response. This lead to development of the FELUDA test for COVID-19 diagnosis. 2)Deployed HTA to assess the innovations 3)testing strategy that used RDP (antigen) and RT-PCR for negative RDT
Pravash Budhathoki, Dhan Bahadur Shrestha, Sitaram Khadka, Santoshi Giri	COVID-19 Status in Nepal and the Way Forward	Europasian Journal of Medical Sciences	review article	Jun-20	Nepal	LMIC	1)Introduction of RDTs in 3 districts 2)PCR testing expanded from national public health lab to 17 centres
Atieme Ogbolosingha	COVID-19 pandemic: Review of impediments to public health measures in SSA	American Journal of Preventive Medicine and Public Health · January 2020	review article	Jun-20	South Africa	UMIC	1) Private pathology labs in SA to supplement public facilities 2) Mobile testing units in SA
					Nigeria	LMIC	1) Mobile testing units
Zheng Li , Cynthia Jones, Girum S. Ejigu, Nisha George, et al	Countries with delayed COVID-19 introduction – characteristics, drivers, gaps and opportunities	Globalisation and Health	research article	2021	24 countries	Mixed	1)Lesotho, Sao Tome & Principle, Cook Islands, Kiribati, Marshal Islands, Micronesia, Nauru, Palua,Samoa, Solomon Islands no incountry testing so export COVID testing was done

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Olivier Vandenberg, Delphine Martiny, Olivier Rochas , Alex van Belkum, Zisis Kozlakidis	Considerations for diagnostic COVID-19 tests	Nature reviews-Microbiology	review article	Oct-20	Several		1)Leveraging platforms used for SARS-CoV and MERS to develop tests 2) government intervention seg financial investments and policies instituted to support manufacturiong capacity 3) USA introduced legislation to alter regulatory frameworks governing lab developed tests 3)continued operation of manufacturing businesese 4) repurposing of manufacturing capacity 5)Centralised system in UK, clinical lab network in Germany, 6) setting up networks of regional sequencing centres hosted in academic and or publi chealth agencies,,
Shabir Ahmad Lone and Aijaz Ahmad	COVID-19 pandemic – an African perspective	Emerging Microbes & Infections	review article	May-20	Several	mixed	1) Continental efforts like AFCOR hel[ped increase testing labs in African from 2 in Feb to 43 by March
Amy S Patteerson and Emmanuel Balogun	African responses to COVID-19: The reckoning of agency	african: studies review	review article	Feb-21	Several		1) Donations of test kit from China 2) Contenental efforts eg Africa CDC developed a COVID-19 Joint Continental Strategy which inlcuded training HCWs in lab dx, 3) PACT formed to stregthen testing capacity on the continent, it procured testing kits and distributed them to several countries, and also created distribution warehouses & distribution hubs 4)AFCOR created btn Africa CDC, WAHO and WHO to increase number og lad conducting testing 5) Partnered with Unitaid, gates foundation, CHAI on devt of rapid diagnostics
Alfonso Medinilla, Bruce Byiers and Philomena Apiko	Africa regional responses to COVID-19	ecdpm making policies work	discussion paper	May-20	Several		1) central role played by Africa CDC in boosting testing capabilities and trainings (it improved testing capacity of labs across the continet, supplying essential equipment to 48 countries and also provided training 2) Throuf PACT, estbalished warehousing and distribution hubs across Africa, coordinated the pooled procurment of diagnostics material, and training CHWs to support contact tracing

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Zachary Desson, Emmi Weller, Peter McMeekin, Mehdi Ammi	An analysis of the policy responses to the COVID-19 pandemic in France, Belgium, and Canada	Health policy and technology	review article	Dec-20	Canada	HIC	1) Initially had one testing centre and gradually capacity was built in other provinces
					France	HIC	1) France massive testing of suspected cases by hospitals, labs, vet clinics and in both govt and private facilities, which led to non harmonised reporting and later special temporary regislation allowed a secure paltform to allow consolidation of results,
					Belgium	HIC	
Luke Taylor	Uruguay is winning against covid-19. This is how	BMJ	feature	Sep-20	Uruguay	HIC	1)Uruguay ramped testing by developing a national testing kit to shield it from market shortages 2) created a network of testing labs in weeks; which included academic labs 3) decentralised testing
Elisabeth Mahase	Covid-19: mass testing in Slovakia may have helped cut infections	BMJ	news	Dec-20	Slovakia	HIC	population wide (mass testing) using an rapid antigen test administered by trained medical peronnel instead of self testinf. Onluy those that tested and were negative were allowed to move
GRID COVID-19 Study Group	Combating the COVID-19 pandemic in a resource-constrained setting: insights from initial response in India	BMJ Global health	Practice article	Oct-20	India	LMIC	1) Increased testing and collection centres 2) expanded testing criteria, 3) engagement of private labs, and existing labs upgraded 4) free of costs testing 5) sample collection kiosks and mobile COVID testing vans 6)(ma6) made in india testing kits on market
Thuong Vu Nguyen, Quang Dai Tran, Lan Trong Phan, Long Ngoc Vu	In the interest of public safety: rapid response to the COVID-19 epidemic in Vietnam	BMJ Global health	Practice article	Jan-21	Vietnam	LMIC	1)collecting samples from people at risk 2) local production of RT-PCR diagnostic kit 3)developed and stocked diagnostic testing products (primers & probes) and antibody tests 4) staff training and assistance 5) increased lab capacity from 2 to 120 labs in 8 months

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Senjuti Saha, Madhukar Pai	Can COVID-19 innovations and systems help low- and middle-income countries to re-imagine healthcare delivery?	Med	Commentary	Apr-21	Bangladesh	LMIC	1) Opened up multiple testing labs 2)multiple groups of people within any basic lab training asked to volunteer & support lab techs 3) govt & lab offered training on sample collection across the nation 4) local reagent & supply vendors solved supply chain puzzles to import supplies from different parts 5) expanded program of immunisation and local WHO office leveraged expertise & network of distributing vaccines to instead transport C-19 samples 6) local sequencing capacity and did sequencing for C-19
					India	LMIC	1) utilisation of existing technologies & resources and repurposing existing technologies
Sangchun Choi, Chul Han, Jisook Lee, Sang-Il Kim, In Byung Kim	Innovative screening tests for COVID-19 in South Korea	Clinical and Experimental Emergency Medicine	Commentary	Apr-20	South Korea	HIC	1) established drive through testing for C-19, 2) testing for C-19 at a walking-through site
Patrick Mbulaje , Steven Munharo, Symon F. Nayupe, Parth Patel	Challenges of COVID-19 testing in low-middle income countries (LMICs): the case of Malawi	Journal of laboratory and precision medicine	letter to the editor	Sep-20	Malawi	LIC	1) with support from UNICEF an existing RT-PCR machine at Lilingwe NRL was set up 2) complentary efforts by teaching and research instiutions and Malawi-Liverpool Wellcome Trust 3) increased the number of testing facilities in the country 4)
Jean B. Nachega, Ashraf Grimwood, Hassan Mahomed, Geoffrey Fatti, et al	From Easing Lockdowns to Scaling Up Community-based Coronavirus Disease 2019 Screening, Testing, and Contact Tracing in Africa—	Clinical infectious disease	view point	Jan-21	South Africa	UMIC	1) deployed CHCWs for screening & referral for testing 2) leveraging UBV infrastructure for COVID-19, and other diseases like malaria, TB,HIV
					DRC	LIC	1) mplementation of multiprofessional health system of nurses, drs, pharmacists, medical students &CHCWS in senitisation, screening, and testing referral
					Rwanda	LIC	1) Sample collection and referral for testing

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	Shared Approaches, Innovations, and Challenges to Minimize Morbidity and Mortality						
Nancy Odendaal	Constructing an "infrastructure of care" – understanding the institutional remnants and socio-technical practices that constitute South Africa's Covid-19 response	Urban geography	view point	Jul-20	SA	UMIC	1) community wide testing using CHWs in areas identified thru a social vulnerability index
H. Sibiri, S.M. Zankawah, D. Prah	Coronavirus diseases 2019 (COVID-19) response: Highlights of Ghana's scientific and technological innovativeness and breakthroughs	ethics medicine and public health	commentary	May-20	Ghana	LMIC	1) pooling techinique 2) dron etechnology to transport samples 3)COVID-19 tracker up to identify who needs to be tested 4)development of RDTs
Jimoh Amzat, Kafayat Aminu, Victor I. Kolo, Maryann C. Danjibo	Coronavirus outbreak in Nigeria: Burden and socio-medical response during the first 100 days	internationa ljournal of infectious diseases		Sep-20	Nigeria	LMIC	1) prioritised those that need to be tested-returnees, symptomatic, contacts, those in high prevalence areas 2) trainingexperts in RT-PCR by Africa CDC 3) Increased number of testing labs from 5 to 23 within 6 months - only public minus private were testing
A. Waris, U.K. Atta, M. Ali, A. Asmat, A. Baset	COVID-19 outbreak: current scenario of Pakistan	new microbes and new infections	mini review	May 202	Pakistan	LMIC	1)Increased number of testing labs 2) national institute of health opened up a mobile testing lab 3) NIH partnered with national disaster management authority to launch training

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							program for paramdecis and lab staff to overcome HR issues
Ryenchindorj Erkhembayar, Emma Dickinson, Darmaa Badarch, Indermohan Narula et al	Early policy actions and emergency response to the COVID- 19 pandemic in Mongolia: experiences and challenges	The Lancet Global Health	health policy	Sep-20	Mongolia	LMIC	1)ordered for probes, primers and possitve controls from Japan 2) used already available platforms 3) lab testing organised by national influenza centre virology lab 4) random community samopliong and walk in testing
Adugna Abera, Habtamu Belaya, Aboma Zewudea, Bokretsion Gideya, et al	Establishment of COVID-19 testing laboratory in resource-limited settings: challenges and prospects reported from Ethiopia	GLOBAL HEALTH ACTION	narrative	Oct-20	Ethiopia	LIC	1)Led by the EPHI, the malaria and NTD research lab used platforms originally used for other diseases eg ifluenza& shcistosomisais to set up the 1st testing lab 2) Training of lab personnel by the incountry virology experts
Santosh Dhakal *and Surendra Karki*	Early Epidemiological Features of COVID-19 in Nepal and Public Health Response	frontiers in medicine	original research	Aug-20	Nepal	LMIC	1)increased number of testing labs from 1 to 20 in a short period 2) included veterinary labs
Saeed Anwar, Mohammad Nasrullah and Mohammad Jakir Hosen	COVID-19 and Bangladesh: Challenges and How to Address Them	frontiers in medicine	policy brief	Apr-20	Bangladesh	LMIC	1) locally developed rapid diagnostic 2) expansion of labs from one to 17 including academic institution

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Ahsan Nawaz, Xing Su, Muhammad Qasim Barkat, Sana Asghar, et al.	Epidemic Spread and Its Management Through Governance and Leadership Response Influencing the Arising Challenges Around COVID-19 in Pakistan—A Lesson Learnt for Low Income Countries With Limited Resource	frontiers in medicine	narrative review	Dec-20	Pakistan	LMIC	1) increased labs precoid to 57 labs after covid 2) training program for staff to increase number ofo techs and experts in molecular biology
Nega Assefa, Jemal Yousuf Hassen, Desalegn Admassu, Mussie Brhane et al	Existing Facilities in Public Health Emergency Management	frontiers in medicine	case study	Jun-21	Ethiopa	LIC	1) support and gyidance from EPHI to academic institutions to set up testing in the academic institution. Support inlcuded training, protocols among others
Percy Herrera-Añazco, Angela Uyen-Cateriano, Edward Mezones-Holguin, Alvaro Taype-Rondan et al.	Some lessons that Peru did not learn before the second wave of COVID-19	international journal of health management	letter to the editor	2021	Peru	UMIC	1)deployed massive use of rapid serological tests as diagnostic and epidemiological tools
Jaroslav Frnda and Marek Durica	On Pilot Massive COVID-19 Testing by Antigen Tests in Europe. Case Study: Slovakia	infectious disease reports	research paper	Jan-21	Slovakia	HIC	Population/country wide rapid antigen testing

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Azizah F. Siddiqui, Manuel Wiederkehr, Liudmila Rozanova and Antoine Flahault	Situation of India in the COVID-19 Pandemic: India's Initial Pandemic Experience	International journal of Environmental research and public health	case report	Dec-20	India	LMIC	1)gradual increase in number of testing labs from 1 to 11 to 885 by end of June 2) private labs included too
Ki Tae Kwon, Jae-Hoon Ko, Heejun Shin, Minki Sung and Jin Yong Kim	Drive-Through Screening Center for COVID-19: a Safe and Efficient Screening System against Massive Community Outbreak	infectious diseases, microbiology and parasititology	brief communication	Mar-20	Korea	HIC	1)introduction of drive through screening and testig centres
Tafadzwa Dzinamarira, Mathias Dzobo, Itai Chitungo	COVID-19: A perspective on Africa's capacity and response	journal of medical virology	review	Jun-20	Nigeria	LMIC	1) Leveraged the existing health infrastructure programs used for previous poliovirus
					Sierra Leone	LIC	1) In Sierra Leone they leveraged structures previoulsy deployed for ebola
Ghulam N Dhabaan, Walid A Al-Soneidor, Nezar N Al-Hebshi	Challenges to testing covd-19 in conflict zones: Yemen as an example	journal of global health	view point	Jun-20	Yemen	LIC	1)used 4 of the 6 central public health labs to test, but this was largely supported by WHO (who trained and also gave supplies)
Al-Musthapha AI, Tijani AA, Oyewo M, Ibrahim A, Elelu N, et al	Nigeria's race to zero COVID-19 cases: True disease burden or testing failure?	Journal of Global Health	view point	Aug-21	Nigeria	LMIC	1) increased labs from the initial 6 to 124 in 4 months 2) leveraged the exisiting platforms used for endemic and sporadic outbreaks 3) deployed xpert platform previoulsy used for HIV and TB 4) private and corporate labs included in testing
Alberto Boretti	The Increasing Number of Infected in Victoria,	integrative journal of medical sciences	case report	Aug-20	Australia	HIC	1)identification of hotspots and mass testing including door to door testing

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	Australia since June 15, 2020, is the Result of Over-Testing and Over-Controlling without Safety						
Godfred Acheampong, Michael Owusu, Bernard Nkrumah, Priscilla Obeng-Boadi, et al.	Laboratory capacity in COVID-19 diagnosis and the need to enhance molecular testing in Ghana	global security: health, science and policy	research article	Mar-21	Ghana	LMIC	1)initially used 3 of the 4 public health labs that had capcity and also used 2 of the acadmic institutions that had capacity 2)gradual increase in number of labs including non clinical labs eg vet labs and academic labs 3) accredited private labs 4)leveraged the GeneXpert platforms 5) scientists among first to sequence the sars-cov-2
Karen E. Ferree, Adam S. Harris, Boniface Dulani, Kristen Kao, et al.	Stigma, Trust, and procedural integrity: Covid-19 testing in Malawi	world development	research article	Jan-21	Malawi	LIC	1) increased labs from 1 to 11 in around 2.5 months, 2)focus on suspected cases , immediate contacts and travelers into the country
María Alejandra Benítez, Carolina Velasco, Ana Rita Sequeira, Josefa Henríquez et al.	Responses to COVID-19 in five Latin American countries	Health policy and technology	narrative review	Aug-20	Chile	HIC	2) Adopted pooled testing in elderly homes 2) introduced low cost tests for COVID
					Peru	UMIC	1)they increased labs doing testing gradually
Peter J. Chipimo, Danielle T. Barradas, Nkomba Kayeyi, Paul M. Zulu, et al.	First 100 Persons with COVID-19 — Zambia, March 18–April 28, 2020	morbidity and mortality weekly report	report	Oct-20	Zambia	LMIC	1)expanded national testin gstrategy to include asymptomatic persons with possibel COVID-19 exposures and no international travel
Tinashe K. Nyazika, Rabelani Kaela, Mathias Mugoni, Kudakwashe	Implementation of Antibody Rapid Diagnostic Testing versus Real-Time	American society for microbiology	perspective	Jul-20	Ghana	LMIC	1) Deployed sample pooling later in the pandemic
					Malawi	LIC	1) Increased testing coverage by increasing number of testing labs

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Musomekwa, et al.	Reverse Transcription-PCR Sample Pooling in the Screening of COVID-19: a Case of Different Testing Strategies in Africa				South Africa	UMIC	1) Deployed targeted community symptom screening and testing for both symptomatic and asymptomatic individuals in some provinces
					Zimbabwe	LMIC	1) ntroduced RDT as a screening test strategy to complement RT-PCR
Bruce J. Tromberg, Tara A. Schwetz, Eliseo J. Pérez-Stable, Richard J. Hodes, et al.	Rapid Scaling Up of Covid-19 Diagnostic Testing in the United States — The NIH RADx Initiative	New England journal of medicine	special report	Sep-20	USA	HIC	1)Initiatives like RADx to sup port the development, production scale-up, and deployment of accurate, rapid tests across the country
Oppah Kuguyo, Andre Pascal Kengne, and Collet Dandara	Singapore COVID-19 Pandemic Response as a Successful Model Framework for Low-Resource Health Care Settings in Africa?	Integrative biology	Narrative	2020	Singapore	HIC	1)Drew lessons from the sars-cov-1 outbreak in 2002 2) in country biotechnology companies produced several tests
Adisu Kebede, Betty Lanyero, Berhane Beyene, Mayur Lalji Mandalia, et al.	Expanding molecular diagnostic capacity for COVID-19 in Ethiopia: operational implications, challenges and lessons learnt	PanAfrican medical journal	Perspective	Jan-21	Ethiopia	LIC	1)identification and assesssment ot public and private labs with support from EPHI, WHO etc 2) on iste bench training, onsite mentorship and drills for the lab personnel 3) calibration of existing PCR machines to be able to do covid testing using kits donated 4) progressive expansion of network by adding more labs 5) used the national influenza, arbovirus and VHF reference lab at EPHI to estbalish covid testing in the country 6) inclusion of research, academic and private labs in testing

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Matthew M Kavanagh, Ngozi A Erondu, Oyewale Tomori, Victor J Dzau,et al.	Access to life-saving medical resources for African countries:	Lancet	View point		Several		1)Africa CDC established pooled purchasing platform
Paul Adepoju	Africa's struggle with inadequate COVID-19 testing	Lancet microbe	News	May-20	Ghana, Nigeria, South Africa	LMIC	4) Drone delivery service stsarted to transport samples,
					Nigeria	LMIC	1)Involved already existing viral hemorrhagic fevers lab network & other labs with molecular diagnostic capacity to increase coverage 2) health start up LifeBank established covid-19 mobile testing centres
					South Africa	UMIC	2) Involved the private sector in testing
Pascale Ondoa, Yenew Kebede, Marguerite Massinga Loembe, Jinal N Bhiman, et al.	COVID-19 testing in Africa: lessons learnt	Lancet microbe	Comment	Jul-20	Several		1)continental efforts coordinated by AFTCOR included teaming up with SA national institute for ID, Senegalese Institute Pasteur Daker & WAHO to train expert staff from refe labs in Africa.2)Africa CDC Pathogen genomic intelligence institute part of joint ocntinental covid-19 startegy to boost sequecning capcity of 16 AU member states 3)Nigeria activated covid-19 testing at 26 sites using HIV molecular testing and TB xpert instruments 4)Ethiopia reconfigured exisiting paltforms 5)Ethiopia involved academic and animal health labs 6) PACT formed to increase testing and organising AU states as on elarge customer to be able to compete
Lincoln Leehang Lau, Natalee Hung and Kendall Wilson	COVID-19 response strategies: considering inequalities	International journal for equity in health	Commentary	Aug-20	Phillipines	LMIC	1)reconfiguring TB testing facilities to meet testing demands

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	between and within countries						
Olivier Vandenberg, Delphine Martiny, Olivier Rochas, Alex van Belkum *and Zisis Kozlakidis*	Considerations for diagnostic COVID-19 tests	Nature reviews Microbiology	review	Mar-21	USA	HIC	1)introduction of legislation to alter the regulatory framework governin glab developed tests, 2) active scouting and import og resources outside usual territories, 3) repurpoing manufacturing capacity
Edward Kim	Drawing on Israel's Experience Organizing Volunteers to Operationalize Drive-Through Coronavirus Testing Centers	Disaster Medicine and Public Health Preparedness	field report	2020	Israel	HIC	1)operationalised drive through coronavirus testinf centres
Lauran K. Evans, Austin Shinagawa, Sarah Sutton, Lisa Calvo	COVID-19 Drive-Through Point of Screening and Testing (POST) System: A Safe, Efficient, and Adaptable Model for Nasopharyngeal Swab Collection	Disaster Medicine and Public Health Preparedness	original research	2020	USA	HIC	1)implementation of covid-19 drive through point of screening and testing
Anthony Kwabena Sarfo, Shankar Karuppannan	Application of Geospatial Technologies in the COVID-19 Fight of Ghana	Indian National Academy of Engineering	original research	Jul-20	Ghana	LMIC	1)deployed drone technology to help with sample transport

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David Lee and Jaehong Lee	Testing on the move: South Korea's rapid response to the COVID-19 pandemic	Transportation Research Interdisciplinary Perspectives			South Korea	HIC	1)drive through testing 2)used IT to do surveillance and contact tracing . IT inlcuded mobile phone tracking, credit card transaction and video footage from surveillance cams
Dean T. Jamison, Kin Bing Wu	The East–West Divide in Response to COVID-19	Transportation Research Interdisciplinary Perspectives	perspective	Jun-21	Several (China, USA, Vietnam, Europe		1)commisoning of private biotech firms and academic isntitutions to undertake genomic analysis that lead to identificatiomn of the genome hence allowing rapid development of of PCR tests 2)Germany and Korea among first countries to develop tests once genome was shared by China 3)Vietnamm developed itw own kits
PatriciaJ.Garcia, AlexAlarco´n, AngelaBayer, PauloBuss, et al.	COVID-19 Response in Latin America	American journal of tropical medicine and hygiene	perspective piece	Sep-20	Chile	HIC	1) lab screening capacity expanded by including all the country's institutions with PCR equipment and supplying reagents to the lab network of universities and institutes
					Peru	UMIC	1) government purchased serologic tests which were validated and swiftly deployed to scale up testing
Ramanan Laxminarayan, Shahid Jameel, and Swarup Sarkar	India's Battle against COVID-19: Progress and Challenges	American journal of tropical medicine and hygiene	editorial	Aug-20	India	LMIC	1) India ramped up testing by rolling out rapid antigen testing 7 months into the pandemic
Thinley Dorji	The Gross National Happiness Framework and the Health System Response to the COVID-19 Pandemic in Bhutan	American journal of tropical medicine and hygiene	perspective piece	Dec-20	Bhutan	LMIC	1) expanded RT-PCR testing from main centre to more gepgrpahically spread centres 2) increased human resource available (reapproproation of human resource) including engagment of students 3)free covid testing for its citizens 4) initiated population testing of all persons older than 10 years

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Mathew Alexander, Lynn Unruh, Andriy Koval and William Belanger	United States response to the COVID-19 pandemic, January–November 2020	Health Economics, Policy and Law	article	Jan-21	USA	HIC	1)authorisation of private and state and local health labs to develop tetst in February (1st commercial test approved in March -3 months after 1st case). 2) enlistment of private labs in testing.
Aditya Shah, Douglas Challener, Aaron J. Tande, Maryam Mahmood et al.	Drive-Through Testing: A Unique, Efficient Method of Collecting Large Volume of Specimens During the SARS-CoV-2 (COVID-19) Pandemic	Foundation for Medical Education and Research	brief report	Jul-20	USA	HIC	1)drive through testing at Mayo Clinic
Marguerite Massinga Loembé, Akhona Tshangela, Stephanie J. Salyer, Jay K. Varma, Ahmed E. Ogwell Ouma and John N. Nkengasong	COVID-19 in Africa: the spread and response	Nature medicine	comment	Jul-20	Several		1) Africa CDC efforts to increase no of labs 2) leveraged network of community health agents used for previosu diseases eg polio 3) Pasteur institute Dakar Senegal evalusated RDTs
Marc Mendelson, Shabir Madhi,	South Africa's coronavirus testing strategy is broken and not fit for purpose: It's time for a change	South African Medical Journal	Guest editorial	Jun-20	South Africa	UMIC	1) Initiated the community screening and testing (CST) program 2) Recruited field workers to support CST
Reports							

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OECD/European Union 2020	How resilient have European health systems been to the COVID-19 crisis?	Health at a Glance: Europe 2020		Jun-20	Several		1) France limited testing to patients with serious conditions at start of pandemic 2) Iceland conducted large scale testing regime early in outbreak 3) Iceland later did voluntary self referral testing regardless of status 4) Lithuania set up mobile points for testing
Janice Thompson and Sinead McMurray	COVID-19:Testing for Sars-CoV-2 in the UK; and the Use of Testing and Contact Tracing in Selected Countries	Northern Ireland assembly		May-20	UK	HIC	1)Early on "containment phase" testing was for the symptomatics after travel and contacts 2) Later moved to delay phase and tested only those requiring admission to hospital 3) Later rolled out to symptomatic NHS staff and their households 4)academic insittuitons involved
Elizabeth Rough	Coronavirus: Testing for Covid-19	House of commons library	Briefing paper	Oct-20	UK	HIC	1)Decentralised appraoch of testing 2) Inclusion of both public and private providers 3) Inclusion of academic and research institutions 4) containment phase 5) delay phose 6) expanded eligibility to those above 65 with symptoms and later to anyone with 5 yrs and above with symptoms 7) developed point of care tests
Department of Health and Social Care	Coronavirus (COVID-19) Scaling up our testing programmes		Report	Apr-20	UK	HIC	1) new testing lans set up 2) inlcuded universities and reseatch institutes 3) support to new innovators of tests

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Ministry of Finance Planning and Economic development	COVID-19 Interventions Report	MoFPED		Oct-20	Uganda	LIC	1)Operationalising testing labs at border points 2) Accreditation of UVRI bu Africa CDC and WHO 3) Used Genexpert at border points
Infectious diseases sociery of America	COVID-19 Prioritization of Diagnostic Testing	Infectious diseases sociery of America		Mar-20	USA	HIC	1) Expanded testing to a wider group of symptomatic patients in March 2020
FDA	South Korea's Response to COVID-19	Center for Devices and Radiological Health			Korea	HIC	1) Korea leveraged structures from previous epidemics like mers, 2) govt designed EUA pathway for tests and 1st test on market in Jan 3)invited private providers and assured them of ready market 4) linked manufactirers with clinical bodies to conduct clinical studies reducing
Nigeria Centre for Disease Control	National strategy to scal eup access to C-19 disease testing in Nigeria			2020	Nigeria		1) leveraged existing disease structures for Malaria, TB and HIV, 2) built capcity by increasing labs 3) developed a national strategy to scal eup access to c-19testing
Mathieu Ductatel and Francois Godement	Fighting COVID-19: East Asian Responses to the pandemic		Policy paper	Apr-20	North Korea	HIC	1) Started developing very quickly the NAA tests 2) Also started developing serological tests 3) Fast approval processfor medical devices including tests 3) 4) mass testing 7) offering PCR tests offered free of charge

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					China	HIC	1) started developing very quickly the NAA tests 2) Also started developing serological tests 3) Fast approval processfor medical devices including tests
					Singapore	HIC	1) PP partnerships to develop/produce tests
Congressional Research service	COVID-19 testing: Key issues		In focus	Apr-20	USA	HIC	1) initially used public health labs but later included commercial and clinicial labs in hospitals and universities, 2) FDA allowed labs and commercial manufacturers ti use and market test kits prior to receiving EUA
Koichi Kameda, Mady Malheiros Barbeitas, Rosangela Caetano, Ilana Lowy et al.	Testing COVID-19 in Brazil:fragmented efforts and challenges to expand diagnostic capcity at the Brazilian Unified Nationak Health system	Reports in Public health	bried communication	Feb-21	Brazil	UMIC	1)initially testing in hospital testing for symptomatics 2) Included public labs and labs from public universities 2) leveraged HIV/AIDs viral load testing labs 3) increased labs testing for covid 4) public private partnerships 5) private developed leveraged molecualr system developed for blood screening to test covid 6) fast tracking market authorisation of COVID tets by EUA
Pan African Health Organisation	Pan American Health Organisation Response to COVID-19 in the Americas		report	May-20	Sevral		1) ordered and provided test kits to member states 2) conducted trainings 3) leveraged the regions nationak influenza centred and severe acute respiratory infection lab network

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Theme	Definition	When to use	When not to use
Accessibility	The interventions include approaches deployed to make the tests affordable and accessible to the population	Apply this code to mean strategies that brought testing closer to the people, covered more people, in addition to making testing affordable. Interventions that fall under this could include increasing number of testing facilities; categories of individuals tested among others.	Do not use this code if interventions are proposed, but have not been implemented
Diagnostics	This covers all the diagnostic assays available and approaches to ensuring they are in plenty	Apply this code to capture all strategies that resulted in new or more testing assays on the market. This could include use of both PCR and RDTs, repurposing manufacturing plants for manufacturing of diagnostics among others	Do not use this code if there is no evidence that this indeed happened.
Regulatory & Legislation	Regulatory and legislative frameworks that allowed several players onto the market to help boost availability of testing	Apply this code where there is indication that existing legislation and regulatory frameworks were made more accommodative and flexible to handle the immediate challenges around COVID-19 testing. Use if there is an indication that the red tape was reduced.	Do not use this code if the frameworks were more prohibitive than accommodative
Integration	Integration of COVID-19 testing in the existing health care systems without having to reinvent the wheel	Apply this code to descriptions on where countries used the already existing systems to increase testing. This could include systems available for the other diseases e.g., Malaria, TB, HIV among others.	Do not apply this code if there is no evidence that these systems were actually leveraged. Do not include if these were mere proposals, and did not happen.

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Personnel	The number of personnel available to do COVID-19 testing as well as approaches that improve capacity of the personnel to do COVID-19 testing	Apply this code to strategies that involved increasing capacity and number of for the human resources.	Do not apply this code to criteria that does not relate to increasing number of personnel available for COVID-19 testing. Also do not include if there is no indication of such trainings
Technology & Innovation	Interventions that leverage technology and also use the available testing systems more efficiently.	Apply this code in instances where innovative ways to increase testing were adopted. This could include better use of technology and testing systems., as well as efficient testing approaches.	Do not apply this code if the routine or expected is what has been done
Collaboration	Countries develop partnerships both in country and out of the country to help with increased testing. These partnerships are aimed at supporting countries to expand testing.	Apply this code to instances where countries created partnerships to help with increased testing. These partnerships could be aimed at increasing testing equipment, reagents, trainings, guidelines among others	Do not use this code if partnerships or collaborations did not lead to influence testing

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	Strategy	Countries	Thematic area (6)
1	Increase in no. of labs	Bangladesh, Malawi, Nepal, Pakistan, Yemen, Nigeria, Ghana, Malawi, Peru, Ethiopia, Chile, Bhutan, UK, Uganda, Nepal, Canada, Uruguay, Vietnam, Nigeria, Brazil	Accessibility
2	Health technology assessments	India	Accessibility
3	Testing strategy to incorporate use of RDTs	India, Zimbabwe, Peru	Diagnostics
4	Licensing private providers/labs in testing	South Africa, India, Nigeria, Ghana, Ethiopia, China, Chile, USA, UK, North Korea, Singapore, Brazil	Regulatory & Legislation
5	Allowing research and academic institutions	South Africa, Nigeria, France, Uruguay, Malawi, Ethiopia, Ghana, Chile, UK, USA, South Korea, Singapore, Brazil	Regulatory & Legislation
6	Setting up mobile testing units	India, Nigeria, South Africa, Pakistan, Lithuania	Accessibility
7	Leveraging existing platforms for testing diseases	India, Malawi, South Africa, Mongolia, Ethiopia, Nigeria, Ghana, Philippines, Uganda, Brazil	Integration
8	Leveraging infrastructure of other disease programs	Bangladesh, South Africa, Mongolia, Nigeria, Sierra Leone, Ethiopia, Uganda	Integration
9	Development of molecular tests	Uruguay, Vietnam, Singapore, China, South Korea, Germany, USA	Diagnostics
10	Development and subsequent use of RDTs	India, Ghana, Bangladesh, Chile, Vietnam, UK, South Korea, China	Diagnostics
11	Repurposing manufacturing capacity	USA	Diagnostics
12	Training of lab people	Vietnam, Bangladesh, Nigeria, Pakistan, Ethiopia	Personnel
13	Increased HR base	Bangladesh, DRC, Bhutan, South Africa	Personnel
14	Legislation to allow for development of tests	USA, South Korea, China, Brazil	Regulatory & Legislation
15	Population wide screening/mass testing	South Africa	Accessibility
16	Expanding categories of people to test	Belgium, India, Nigeria, Pakistan, Ghana, Zambia, Bhutan, Iceland, UK, USA	Accessibility
17	Testing only symptomatic and high priority groups	Vietnam, Nigeria, Malawi, France, UK, Brazil	Accessibility
18	Pooled testing	Ghana, Chile	Technology & Innovation
19	Drive through testing/walk in testing	South Korea, Israel, USA	Accessibility
20	Community wide testing using CHWs	South Africa, Mongolia	Accessibility
21	Sample collection and referral of samples	India, DRC, Rwanda	Accessibility
22	Use of veterinary labs	France, Nepal	Integration
23	Massive testing with RDTs	Slovakia, Peru	Diagnostics
24	Targeted/hot spot with mass testing	Slovakia, Australia, South Africa	Accessibility

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25	Initiatives to support production of diagnostics	USA, South Korea, China	Diagnostics
26	Free COVID testing	India, Bhutan, South Korea	Accessibility
27	IT to do surveillance and contact tracing	Ghana, South Korea	Technology & Innovation
29	Drone technology to ship samples	Ghana	Technology & Innovation
20	Continental led efforts (Africa CDC)- pooled purchasing, capacity to country labs, training	Africa, South America, North America by Africa CDC and PAHO	Collaboration

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	3
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	6,7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	8
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	7
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	NA
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	NA
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	NA
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NA
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	NA
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	9
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	9
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NA
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	9
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	26,27
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	NA
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	NA
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	NA
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	18
	23b	Discuss any limitations of the evidence included in the review.	18
	23c	Discuss any limitations of the review processes used.	18
	23d	Discuss implications of the results for practice, policy, and future research.	18
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Registration started but not completed
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Protocol not prepared
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	19
Competing interests	26	Declare any competing interests of review authors.	19
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	19

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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